JEE Mathematics: Quick Review with Grading and Practice Map - Vector 3D (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (292 questions; a question that appears in two of these is counted once): High = 27 or more questions, Medium = 1 to 26 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction

Question 01

Define vector, initial point, terminal point, head and tail of a vector

Model Answer

A vector is a directed line segment (magnitude + direction, obeys vector addition). In $\overrightarrow{AB}$, A = initial point (tail), B = terminal point (head).
$\overrightarrow{AB} \neq \overrightarrow{BA}$: the end points cannot be interchanged; $\overrightarrow{BA} = -\overrightarrow{AB}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

What is the direction of a vector?

Model Answer

The direction from the tail to the head along its line of support; it is given by the unit vector $\hat{a} = \dfrac{\vec{a}}{|\vec{a}|}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 03

What is the magnitude of a vector?

Model Answer

The length of the directed segment: $|\overrightarrow{AB}| = AB \geq 0$, and $|\overrightarrow{AB}| = |\overrightarrow{BA}|$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-II 7

Question 04

What is line of support of a vector?

Model Answer

The line of unlimited length of which the directed segment is a part (also called the support).

Grade - Low

Practice map, main concept: none

Also used in: none

Important Definitions

Question 05

Define null vector and how is it represented?

Model Answer

A vector of magnitude zero, written $\vec{0}$; initial and terminal points coincide, its direction is arbitrary.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 06

Define unit vector and how is it represented?

Model Answer

A vector of magnitude 1. Unit vector along $\vec{a}$: $\hat{a} = \dfrac{\vec{a}}{|\vec{a}|}$; $\hat{i}, \hat{j}, \hat{k}$ are along the axes.

Ex: $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k} \Rightarrow \hat{a} = \frac{1}{3}(2\hat{i} - \hat{j} + 2\hat{k})$.

Grade - Medium

Practice map, main concept: Question Bank: 37

Also used in: Sheet: J-I 1, 8; J-II 53, 56; J-M 159, 161 | Question Bank: 1, 33

Question 07

How to find magnitude of a vector expressed in $\hat{i}, \hat{j}, \hat{k}$ form?

Model Answer

$$|x\hat{i} + y\hat{j} + z\hat{k}| = \sqrt{x^2 + y^2 + z^2}$$

Ex: $|3\hat{i} - 4\hat{j} + 12\hat{k}| = 13$. For points: $AB = |\vec{b} - \vec{a}|$.

Grade - Medium

Practice map, main concept: Sheet: J-M 71 | DPP: 1(3)

Also used in: Sheet: J-A 48; C-I 15; C-II 28 | DPP: 1(6); 3(6); 9(8); 10(12) | Question Bank: 32, 50

Question 08

What is the role of unit vector?

Model Answer

It gives only the direction. A vector of magnitude k along $\vec{a}$ is $k\hat{a}$ (opposite: $-k\hat{a}$).

Ex: magnitude 6 along $2\hat{i} - \hat{j} + 2\hat{k}$: $4\hat{i} - 2\hat{j} + 4\hat{k}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 56; J-M 172

Also used in: Sheet: J-I 26; J-II 15

Question 09

When are two vectors equal?

Model Answer

When they have the same magnitude and the same direction (their position in space does not matter).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 10
How to equate two vectors expressed in $\hat{i}, \hat{j}, \hat{k}$ form?

Model Answer
Compare the coefficients: $a_1 = b_1,\ a_2 = b_2,\ a_3 = b_3$.
Ex: $(x+y)\hat{i} + (x-y)\hat{j} + z\hat{k} = 5\hat{i} + \hat{j} - 2\hat{k} \Rightarrow x = 3,\ y = 2,\ z = -2$.

Grade - Medium
Practice map, main concept: Sheet: C-II 18, 22
Also used in: Sheet: J-I 22; J-II 10 | Question Bank: 24

Question 11
How is the angle between two vectors defined and how is angle denoted?

Model Answer
The smaller angle θ formed when the tails (or the heads) of the two vectors are brought together; written $\theta = (\widehat{\vec{a}, \vec{b}})$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 12
What is the range of angle between two vectors?

Model Answer

$$0 \leq \theta \leq \pi$$

$\theta = 0$: like parallel; $\theta = \pi$: unlike parallel; $\theta = \frac{\pi}{2}$: perpendicular.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 13
How to calculate angle if head and tail are placed together?

Model Answer
Shift one vector so that both start from the same point; if the figure angle is α between head-to-tail vectors, the required angle is $\pi - \alpha$.
Ex: equilateral $\triangle ABC$: angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$ is $120°$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 14
What do you mean by collinear vectors?

Model Answer
Vectors whose lines of support are parallel or the same (also called parallel vectors), whatever their directions.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 15
How to check whether two collinear vectors are like or unlike vectors?

Model Answer
Write $\vec{a} = \lambda\vec{b}$: $\lambda > 0$ like (same direction), $\lambda < 0$ unlike (opposite).
Ex: $2\hat{i} - 4\hat{j} + 6\hat{k} = -2(-\hat{i} + 2\hat{j} - 3\hat{k})$: unlike.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 16
What is the condition of two vectors, expressed in $\hat{i}, \hat{j}, \hat{k}$ form, to be collinear?

Model Answer

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} \ (= \lambda)$$

Points A, B, C are collinear $\Leftrightarrow \overrightarrow{AB} = \lambda\overrightarrow{AC}$.
Ex: $(x+2)\hat{i} - (x-y)\hat{j} + \hat{k} \parallel (x-1)\hat{i} + (2x+y)\hat{j} + 2\hat{k} \Rightarrow x = -5,\ y = -20$.

Grade - Low
Practice map, main concept: Sheet: C-II 25
Also used in: Sheet: J-II 8 | DPP: 1(7)

Question 17
How to assume a vector parallel to a given vector $\vec{p}$?

Model Answer
Take it as $\lambda\vec{p}$, $\lambda \in \mathbb{R}$; then use the extra condition (magnitude, dot product...) to find λ.

Grade - Medium
Practice map, main concept: Sheet: J-I 4; J-M 124 | DPP: 3(6); 10(5)
Also used in: Sheet: J-I 17; J-M 2, 161; J-A 2 | DPP: 4(3) | Question Bank: 25, 55

Question 18
What do you mean by coplanar vectors?

Model Answer
Vectors whose directed segments lie in (or are parallel to) one plane.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 19
How to check coplanarity of two vectors?

Model Answer
No check is needed: any two vectors are always coplanar (bring their tails together). Three vectors need $[\vec{a}\ \vec{b}\ \vec{c}] = 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 20

What to keep in mind while shifting vectors in Mathematics?

Model Answer

Vectors are free: they may be shifted anywhere keeping magnitude and direction unchanged. Localised vectors (e.g. force at a point) are not shifted.

Grade - Low

Practice map, main concept: none

Also used in: none

Addition of vectors

Question 21

How to write relation between vectors forming a closed figure?

Model Answer

Taken head to tail around a closed figure, the sum is $\vec{0}$: $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=\vec{0}$ (triangle law $\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$).

Ex: if C is the mid-point of AB, then $\overrightarrow{PA}+\overrightarrow{PB}=2\overrightarrow{PC}$ for any P.

Grade - Medium

Practice map, main concept: Sheet: J-M 80; C-I 16 | DPP: 2(3); 8(2)

Also used in: Sheet: J-I 10, 15; J-M 78; J-A 18, 25, 36 | DPP: 1(6, 9) | Question Bank: 38, 56

Question 22

How to write diagonal vectors of a parallelogram whose adjacent side vectors are $\vec{a}$ and $\vec{b}$?

Model Answer

$$\vec{d_1}=\vec{a}+\vec{b}, \quad \vec{d_2}=\vec{b}-\vec{a}$$

Also $|\vec{a}+\vec{b}|^2+|\vec{a}-\vec{b}|^2=2(|\vec{a}|^2+|\vec{b}|^2)$.

Grade - Medium

Practice map, main concept: Sheet: C-II 21 | DPP: 3(4); 5(3)

Also used in: Sheet: J-I 6; J-M 129; J-A 6

Question 23

How to add two vectors expressed in $\hat{i},\hat{j},\hat{k}$ form?

Model Answer

Add the corresponding components: $\vec{a}+\vec{b}=(a_1+b_1)\hat{i}+(a_2+b_2)\hat{j}+(a_3+b_3)\hat{k}$.

Ex: $(\hat{i}+2\hat{j}-\hat{k})+(2\hat{i}-\hat{j}+3\hat{k})=3\hat{i}+\hat{j}+2\hat{k}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 24

What do you mean by additive inverse?

Model Answer

$-\vec{a}$: same magnitude, opposite direction, with $\vec{a}+(-\vec{a})=\vec{0}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 25
Write expression of triangle inequality.

Model Answer

$$\big||\vec{a}| - |\vec{b}|\big| \leq |\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$$

Ex: $|\vec{a}| = 3,\ |\vec{b}| = 4 \Rightarrow 1 \leq |\vec{a} + \vec{b}| \leq 7$.

Grade - Low
Practice map, main concept: Sheet: C-II 36
Also used in: none

Question 26
What is the condition on two vectors in case of equality in triangle inequality?

Model Answer
$|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$ iff $\vec{a}, \vec{b}$ are like parallel; $|\vec{a} + \vec{b}| = \big||\vec{a}| - |\vec{b}|\big|$ iff unlike parallel (or one is $\vec{0}$).

Grade - Low
Practice map, main concept: none
Also used in: none

Multiplication of a vector by scalar

Question 27
What is the effect on magnitude of a vector if a vector gets multiplied by λ?

Model Answer

$$|\lambda\vec{a}| = |\lambda|\,|\vec{a}|$$

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 14

Question 28
What is the effect on direction of a vector if a vector gets multiplied by λ?

Model Answer
$\lambda > 0$: same direction; $\lambda < 0$: opposite direction; $\lambda = 0$: null vector.
Ex: $-2\vec{a}$ has twice the length of $\vec{a}$ and the opposite direction.

Grade - Low
Practice map, main concept: none
Also used in: none

Coordinate System in 3D

Question 29
How to define x, y and z coordinates of a point?

Model Answer
x, y, z = signed distances of P from the yz-, zx- and xy-planes; position vector $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$. The three coordinate planes divide space into 8 octants.

Ex: origin shifted to $(1, 2, -3)$ (axes parallel): $(0, 4, 5)$ becomes $(-1, 2, 8)$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 9; J-M 61 | DPP: 8(14)

Question 30
What is the condition on coordinates if a point $P(a, b, c)$ lies on x-axis / y-axis / z-axis?

Model Answer
x-axis: $b = c = 0$; y-axis: $a = c = 0$; z-axis: $a = b = 0$. Planes: xy: $c = 0$, yz: $a = 0$, zx: $b = 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 31
What is the condition on coordinates if $P(a, b, c)$ lies on xy-plane / yz-plane / zx-plane?

Model Answer
xy-plane: $c = 0$; yz-plane: $a = 0$; zx-plane: $b = 0$ (the coordinate missing from the plane's name is zero).

Ex: $(2, 0, -5)$ lies in the zx-plane.

Grade - Medium
Practice map, main concept: none
Also used in: DPP: 6(7); 7(7) | Question Bank: 29

Question 32
What are the coordinates of the foot of perpendicular from $P(a, b, c)$ on x-axis / y-axis / z-axis?

Model Answer
On x-axis: $(a, 0, 0)$; on y-axis: $(0, b, 0)$; on z-axis: $(0, 0, c)$ (keep that axis's coordinate, others zero).

Ex: $P(3, -4, 5)$: foot on y-axis is $(0, -4, 0)$.

Grade - Low
Practice map, main concept: Sheet: J-II 33
Also used in: none

Question 33
What are the coordinates of the foot of perpendicular from $P(a, b, c)$ on xy-plane / yz-plane / zx-plane?

Model Answer
On xy-plane: $(a, b, 0)$; on yz-plane: $(0, b, c)$; on zx-plane: $(a, 0, c)$ (only the missing coordinate becomes zero).

Ex: $P(3, -4, 5)$: foot on zx-plane is $(3, 0, 5)$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 168; J-A 31 | DPP: 6(5) | Question Bank: 36

Question 34
What is the distance of point $P(a, b, c)$ from x-axis / y-axis / z-axis?

Model Answer

$$d_x = \sqrt{b^2 + c^2}, \qquad d_y = \sqrt{c^2 + a^2}, \qquad d_z = \sqrt{a^2 + b^2}$$

(Square root of the sum of squares of the other two coordinates.) Ex: $P(3, 4, 12)$: from x-axis $4\sqrt{10}$, from z-axis 5.

Grade - Medium
Practice map, main concept: Sheet: J-M 168 | Question Bank: 36
Also used in: Sheet: J-A 31; C-I 18

Question 35
What is the distance of point $P(a, b, c)$ from xy-plane / yz-plane / zx-plane?

Model Answer
From xy-plane $|c|$, from yz-plane $|a|$, from zx-plane $|b|$ (modulus of the missing coordinate).
Ex: $P(3, -4, 5)$ is at distance 4 from the zx-plane.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 33

Question 36
How to find new coordinates of a point if origin is shifted to (a, b, c)?

Model Answer

$$x' = x - a, \qquad y' = y - b, \qquad z' = z - c$$

Axes stay parallel to the old axes. Ex: origin to $(1, 2, -3)$: $(0, 4, 5)$ becomes $(-1, 2, 8)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Position Vector and Distance Formula

Question 37
How to find position vector of point $P(a, b, c)$?

Model Answer

$$\overrightarrow{OP} = a\hat{i} + b\hat{j} + c\hat{k}$$

It is the vector from the origin O to P. Ex: $P(2, -1, 3)$: $\overrightarrow{OP} = 2\hat{i} - \hat{j} + 3\hat{k}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 38

How to find coordinates of a point if position vector is given and vice versa?

Model Answer

The coefficients of $\hat{i}, \hat{j}, \hat{k}$ in the position vector are the coordinates, and conversely: $x\hat{i} + y\hat{j} + z\hat{k} \leftrightarrow (x, y, z)$.

Ex: $\vec{r} = 4\hat{i} - 5\hat{k} \leftrightarrow (4, 0, -5)$ (missing component $= 0$).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 39

How to express the vector joining two points A and B in terms of their position vectors / coordinates?

Model Answer

$$\overrightarrow{AB} = \vec{b} - \vec{a} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

(p.v. of head) $-$ (p.v. of tail). Ex: $A(1, -1, 2)$, $B(2, 3, -4)$: $\overrightarrow{AB} = \hat{i} + 4\hat{j} - 6\hat{k}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 2

Also used in: Sheet: J-I 1, 3; J-M 67, 127; J-A 29, 55

Question 40

What is the distance formula in 3D?

Model Answer

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}, \qquad OP = \sqrt{x^2 + y^2 + z^2}$$

Ex: $(1, 2, 3)$ and $(4, 6, 15)$: $AB = \sqrt{9 + 16 + 144} = 13$.

Grade - Medium

Practice map, main concept: Sheet: J-A 49 | DPP: 1(4) | Question Bank: 29

Also used in: Sheet: J-M 30, 34, 114, 117, 145, 151, 158, 181, 184, 194, 197 | DPP: 6(16); 7(7, 10)

Question 41

How to find distance between two points expressed in form of position vectors / coordinates?

Model Answer

$$AB = |\overrightarrow{AB}| = |\vec{b} - \vec{a}|$$

Write $\vec{b} - \vec{a}$ in $\hat{i}, \hat{j}, \hat{k}$ form and take its magnitude (same as the distance formula with coordinates).

Ex: $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 4\hat{k}$: $AB = |\hat{i} + 4\hat{j} - 6\hat{k}| = \sqrt{53}$.

Grade - Medium

Practice map, main concept: DPP: 1(5)

Also used in: Sheet: J-I 2, 41; J-II 58; J-M 121-123, 177; J-A 3, 9, 50; C-I 14; C-II 4-5, 31, 33 | DPP: 1(1) | Question Bank: 8, 13, 17, 27-28, 31, 41, 44-46, 53-54, 56, 61-62, 69

Section Formulae

Question 42

Write the section formula for internal division in form of position vectors / coordinates.

Model Answer

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}, \qquad R\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n}\right)$$

R divides AB internally in $m : n$; mid point $\frac{\vec{a}+\vec{b}}{2}$. Ex: $A(1,2,3)$, $B(4,-1,6)$, $1:2$: $R(2,1,4)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 3; J-II 3-4; J-M 34, 122-123; J-A 55 | DPP: 1(1, 6); 2(7); 6(7) | Question Bank: 20, 45-46

Also used in: Sheet: J-II 48-49, 57; J-M 57, 80, 96, 136; J-A 48; C-II 11, 31-32 | DPP: 2(1, 8); 4(9, 11); 6(17); 9(2) | Question Bank: 51

Question 43

Write the section formula for external division in form of position vectors / coordinates.

Model Answer

$$\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}, \qquad R\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}, \frac{mz_2 - nz_1}{m-n}\right)$$

($m \neq n$; replace n by $-n$ in the internal formula.) Ex: $A(1,2,3)$, $B(4,-1,6)$, $2:1$ externally: $R(7,-4,9)$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 48

Question 44

How to find centroid of a triangle when all three vertices are given?

Model Answer

$$\vec{g} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}, \qquad G\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right)$$

Ex: $(1,2,3)$, $(2,-1,0)$, $(3,5,-6)$: $G(2,2,-1)$.

Grade - Medium

Practice map, main concept: Sheet: C-I 14 | Question Bank: 44, 53-54, 56

Also used in: Sheet: J-II 20-21; J-M 69, 95, 121, 145, 151; J-A 50; C-I 11 | DPP: 1(2); 4(13) | Question Bank: 21

Question 45

How to find incentre / excentres of a triangle when all three vertices are given?

Model Answer

$$\vec{I} = \frac{a\vec{A} + b\vec{B} + c\vec{C}}{a+b+c}, \qquad \vec{I_1} = \frac{-a\vec{A} + b\vec{B} + c\vec{C}}{-a+b+c}$$

$a = BC$, $b = CA$, $c = AB$ (find them by the distance formula); I_2, I_3: change the sign of b, c respectively.

Ex: $A(0,0,1)$, $B(3,0,1)$, $C(0,4,1)$: $a = 5, b = 4, c = 3$; $I(1,1,1)$, $I_1(6,6,1)$.

Grade - Low

Practice map, main concept: none

Also used in: DPP: 1(10); 4(13) | Question Bank: 45-46, 51

Question 46

What is the formula for position vector of circumcentre of a triangle?

Model Answer

$$\vec{O} = \frac{\vec{A}\sin 2A + \vec{B}\sin 2B + \vec{C}\sin 2C}{\sin 2A + \sin 2B + \sin 2C}$$

Ex: right angle at A: circumcentre = mid point of BC; for $A(0,0,1)$, $B(3,0,1)$, $C(0,4,1)$ it is $\left(\frac{3}{2}, 2, 1\right)$.

Grade - Medium

Practice map, main concept: DPP: 4(13)

Also used in: Sheet: J-M 69 | DPP: 1(10)

Question 47

What is the formula for position vector of orthocentre of a triangle?

Model Answer

$$\vec{H} = \frac{\vec{A}\tan A + \vec{B}\tan B + \vec{C}\tan C}{\tan A + \tan B + \tan C}$$

Also $\vec{H} = 3\vec{G} - 2\vec{O}$ (O, G, H collinear, G divides OH in $1:2$). Ex: above triangle: $H = A(0,0,1)$.

Grade - Medium

Practice map, main concept: Sheet: J-M 69, 116, 121; J-A 24; C-II 24 | DPP: 1(2, 10) | Question Bank: 21

Also used in: Sheet: J-II 69 | DPP: 3(7); 4(13) | Question Bank: 44

Collinearity of 3 points

Question 48

How to check whether three given points are collinear?

Model Answer

A, B, C are collinear iff $\overrightarrow{AB} \parallel \overrightarrow{AC}$, i.e. $\overrightarrow{AB} = \lambda\overrightarrow{AC}$ (components proportional), or $\overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$.

Ex: $A(2,3,0)$, $B(p,9,0)$, $C(1,-1,0)$: $\frac{p-2}{-1} = \frac{6}{-4} \Rightarrow p = \frac{7}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 2; J-M 65 | DPP: 1(7)

Also used in: Sheet: J-A 33 | DPP: 2(3)

Question 49

Explain the note on linear relation between position vectors of three collinear points.

Model Answer

Points $\vec{a}, \vec{b}, \vec{c}$ are collinear iff there exist scalars x, y, z, not all zero, with $x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$ and $x + y + z = 0$.

(Coefficient sum zero makes the relation independent of the origin.) Ex: $(x^2 - 2)\vec{a} + 3x\vec{b} + 4\vec{c} = \vec{0}$: $x^2 + 3x + 2 = 0 \Rightarrow x = -1, -2$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 2

Linearly Independent and Dependent Vectors

Question 50

What do you mean by linear combination of vectors?

Model Answer

A vector of the form $x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$ (x_i scalars) is a linear combination of $\vec{a}_1, \ldots, \vec{a}_n$.

Ex: $2\vec{a} - 3\vec{b} + \vec{c}$ is a linear combination of $\vec{a}, \vec{b}, \vec{c}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 51

What do you mean by linearly independent vectors and linearly dependent vectors?

Model Answer

Independent: $k_1\vec{x}_1 + \cdots + k_n\vec{x}_n = \vec{0} \Rightarrow k_1 = \cdots = k_n = 0$; no vector is a linear combination of the others.

Dependent: such a relation holds with some $k_i \neq 0$, so at least one vector is a combination of the others.

Ex: $(\hat{i} + \hat{j}) + (\hat{j} + \hat{k}) - (\hat{i} + 2\hat{j} + \hat{k}) = \vec{0}$, so these three are dependent.

Grade - Low

Practice map, main concept: Sheet: J-II 63

Also used in: none

Question 52

When can we equate coefficients of vectors on both sides?

Model Answer

Only when the vectors on both sides are linearly independent: $\sum x_i\vec{a}_i = \sum y_i\vec{a}_i \Rightarrow x_i = y_i$.

Ex: $\vec{a}, \vec{b}$ non-collinear and $x\vec{a} + y\vec{b} = 2\vec{a} - 3\vec{b} \Rightarrow x = 2,\ y = -3$ (not allowed if $\vec{a} \parallel \vec{b}$).

Grade - Medium

Practice map, main concept: Sheet: J-II 1; C-II 20 | DPP: 4(3, 8)

Also used in: Sheet: J-I 4, 32-33; J-II 3-4; J-M 124; J-A 17; C-II 13 | DPP: 2(8); 10(4-5, 13) | Question Bank: 21

Question 53

When are two vectors linearly independent?

Model Answer

When they are non-zero and non-collinear ($\vec{a} \neq \lambda\vec{b}$, i.e. $\vec{a} \times \vec{b} \neq \vec{0}$). Two collinear vectors are dependent.

Ex: $\hat{i} + 2\hat{j}$ and $2\hat{i} + 4\hat{j}$ are dependent; $\hat{i} + 2\hat{j}$ and $\hat{j}$ are independent.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 1

Question 54
When are three vectors linearly independent?

Model Answer
When they are non-coplanar: $[\vec{a}\ \vec{b}\ \vec{c}] \neq 0$ (determinant of components $\neq 0$). Three coplanar vectors are dependent.
Ex: $\hat{i},\ \hat{i}+\hat{j},\ \hat{i}+\hat{j}+\hat{k}$: determinant $= 1 \neq 0$, independent.

Grade - Low
Practice map, main concept: Sheet: J-I 54 | DPP: 8(9); 9(8); 10(4)
Also used in: Sheet: J-II 66; C-II 20

Question 55
When are more than three vectors linearly independent?

Model Answer
Never (in 3D space): any four or more vectors are always linearly dependent; one of them is a combination of three non-coplanar ones.
Ex: $\hat{i}+\hat{j}+\hat{k}-\hat{i}-\hat{j}-\hat{k}=\vec{0}$ shows $\hat{i},\hat{j},\hat{k},\ \hat{i}+\hat{j}+\hat{k}$ are dependent.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 63

Question 56
Explain Ceva's theorem (concurrent cevians of a triangle) and its vector proof by comparing coefficients.

Model Answer
D, E, F on BC, CA, AB: AD, BE, CF are concurrent iff $\dfrac{BD}{DC}\cdot\dfrac{CE}{EA}\cdot\dfrac{AF}{FB}=1$.
Vector method: take a vertex as origin; write the p.v. of the meeting point from both lines with parameters; as the two side vectors are linearly independent, equate coefficients.
Ex: $BD:DC=3:2$, $CE:EA=1:3$: $\frac{3}{2}\cdot\frac{1}{3}\cdot\frac{AF}{FB}=1 \Rightarrow AF:FB=2:1$.

Grade - Low
Practice map, main concept: none
Also used in: none

Vector Equation of Line

Question 57
How to write vector equation of line and what to keep in mind?

Model Answer

$$\vec{r}=\vec{a}+\lambda\vec{b}$$

$\vec{a}$ = p.v. of any fixed point on the line, $\vec{b}$ = any non-zero vector parallel to it, $\lambda \in \mathbb{R}$; the equation is not unique (another point or a multiple of $\vec{b}$ gives the same line).
Ex: through $(1,2,3)$ parallel to $2\hat{i}-\hat{j}+\hat{k}$: $\vec{r}=\hat{i}+2\hat{j}+3\hat{k}+\lambda(2\hat{i}-\hat{j}+\hat{k})$.

Grade - Medium
Practice map, main concept: Sheet: J-II 49
Also used in: Sheet: J-II 48; J-M 29 | DPP: 3(2); 6(8); 7(11) | Question Bank: 3

Question 58
How to find direction vector of a line passing through 2 points?

Model Answer

$$\vec{b}-\vec{a} \quad \text{(or any non-zero multiple)}, \qquad \vec{r}=\vec{a}+\lambda(\vec{b}-\vec{a})$$

Ex: through $(1,2,3)$ and $(2,0,1)$: direction $\hat{i}-2\hat{j}-2\hat{k}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 50, 58, 100, 105-106, 147, 181, 191-192, 194 | DPP: 5(10) | Question Bank: 41, 69

Question 59
What does $\vec{r}$ denote in equation of a line?

Model Answer

$\vec{r}$ is the position vector of a variable (general) point of the line; each value of λ gives one point of the line.

Ex: $\vec{r}=\hat{i}+\lambda\hat{j}$: $\lambda=2$ gives the point $(1,2,0)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 60
How to assume a point on a line whose equation is known?

Model Answer

$$\vec{a}+\lambda\vec{b} \;\leftrightarrow\; (a_1+\lambda b_1,\; a_2+\lambda b_2,\; a_3+\lambda b_3)$$

Ex: $\vec{r}=(1,2,3)+\lambda(2,-1,1)$: general point $(1+2\lambda,\; 2-\lambda,\; 3+\lambda)$.

Grade - Medium

Practice map, main concept: Sheet: J-M 151, 158, 181 | Question Bank: 41, 69

Also used in: Sheet: J-I 35, 43; J-II 25-26, 51; J-M 30, 39, 142, 156; J-A 8, 32-33; C-I 17; C-II 4 | DPP: 7(2) | Question Bank: 11, 13, 17, 26, 30, 42, 54, 67

Question 61
How to find direction vector of a line whose equation is known?

Model Answer

Write the equation as $\vec{r}=\vec{a}+\lambda\vec{b}$; the vector multiplying the parameter, $\vec{b}$, is the direction vector.

Ex: $\vec{r}=(1+2\lambda)\hat{i}+(2-\lambda)\hat{j}+3\hat{k}$: direction $2\hat{i}-\hat{j}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 170; C-I 20

Classification of line

Question 62

How to find point of intersection of two intersecting lines whose equations are given?

Model Answer

Equate the general points $\vec{a}_1 + \lambda\vec{b}_1 = \vec{a}_2 + \mu\vec{b}_2$; solve two component equations for λ, μ, and put λ back in L_1.

Ex: $(1,2,3) + \lambda(1,1,1)$ and $(2,0,1) + \mu(0,2,2)$: $\lambda = 1,\ \mu = \frac{3}{2}$, point $(2,3,4)$.

Grade - Medium

Practice map, main concept: Sheet: J-M 33, 96, 145, 184; J-A 11 | DPP: 2(6, 8); 4(4) | Question Bank: 28

Also used in: Sheet: J-I 12; J-II 24, 30; J-M 83, 152, 182; J-A 9, 39, 53 | Question Bank: 14, 30-31, 53, 57, 61

Question 63

How to check whether lines are intersecting or skew if their equations are known?

Model Answer

(Non-parallel lines.) Solve two component equations for λ, μ; if the third is satisfied the lines intersect, otherwise they are skew.

Equivalent test: intersecting $\iff (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$.

Ex: $(1,2,3) + \lambda(1,1,1)$, $(2,0,0) + \mu(0,2,2)$: $\lambda = 1, \mu = \frac{3}{2}$ give z: $4 \neq 3$, skew.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 39; J-II 24, 67; J-M 111; J-A 10, 45; C-I 22 | DPP: 1(8); 4(4); 7(1)

Question 64

How to check whether lines are parallel if their equations are known?

Model Answer

Parallel iff the direction vectors are collinear: $\vec{b}_1 = k\vec{b}_2$, i.e. $\dfrac{b_{11}}{b_{21}} = \dfrac{b_{12}}{b_{22}} = \dfrac{b_{13}}{b_{23}}$.

Ex: $\vec{r} = \lambda(3,-1,2)$ and $\vec{r} = (2,0,1) + \mu(6,-2,4)$: $\frac{3}{6} = \frac{-1}{-2} = \frac{2}{4}$, parallel.

Grade - Low

Practice map, main concept: Sheet: J-II 24

Also used in: DPP: 6(11)

Question 65

How to check whether parallel lines are coincident or disjoint if their equations are known?

Model Answer

Check whether a point of L_1 lies on L_2, i.e. $\vec{a}_2 - \vec{a}_1 \parallel \vec{b}$: yes = coincident (infinitely many common points), no = disjoint.

Ex: $\lambda(3,-1,2)$, $(2,0,1) + \mu(6,-2,4)$: $(2,0,1) \nparallel (3,-1,2)$, disjoint. $(0,0,1) + \lambda(3,2,1)$, $(3,2,2) + \mu(6,4,2)$: difference $(3,2,1)$, coincident.

Grade - Low

Practice map, main concept: Sheet: J-II 67 | DPP: 6(11)

Also used in: none

Question 66
How to check whether two equations expressed differently represent the same line?

Model Answer
Both conditions: direction vectors parallel ($\vec{b}_1 \parallel \vec{b}_2$) and the fixed point of one satisfies the other ($\vec{a}_2 - \vec{a}_1 \parallel \vec{b}_1$).
Ex: $(1,0,2)+\lambda(1,-1,2)$ and $(3,-2,6)+\mu(-2,2,-4)$: $(2,-2,4) \parallel (1,-1,2)$, same line.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 67 | DPP: 7(3)

Vector Equation of Angle Bisector

Question 67
How to find vector in direction of angle bisectors of two lines whose direction vectors are known?

Model Answer

$$\text{bisectors: } \lambda(\hat{a}+\hat{b}) \text{ and } \lambda(\hat{a}-\hat{b})$$

Use unit vectors (not $\vec{a}+\vec{b}$ unless $|\vec{a}|=|\vec{b}|$); $\hat{a}+\hat{b}$ bisects the angle between $\vec{a},\vec{b}$, $\hat{a}-\hat{b}$ the exterior angle; the two are perpendicular.
Ex: $\vec{a}=3\hat{i}+4\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}+2\hat{k}$: along $7\hat{i}+5\hat{j}+11\hat{k}$ and $2\hat{i}-5\hat{j}+\hat{k}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 1, 22; J-II 48; J-M 68; J-A 39 | DPP: 9(3) | Question Bank: 23
Also used in: Sheet: J-A 38; C-II 27

Dot Product

Question 68
Write expression of $\vec{a}\cdot\vec{b}$.

Model Answer

$$\vec{a}\cdot\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta, \qquad 0 \le \theta \le \pi$$

θ = angle between $\vec{a}$ and $\vec{b}$ (tails together); the result is a scalar.

Grade - Medium
Practice map, main concept: Question Bank: 5
Also used in: Sheet: J-I 25, 29; J-M 22, 60, 84, 120, 132, 175; J-A 14 | DPP: 2(7) | Question Bank: 16, 19, 43

Question 69
When is dot product of two vectors positive / zero / negative?

Model Answer
Positive: θ acute ($0 \le \theta < \frac{\pi}{2}$); zero: $\theta = \frac{\pi}{2}$ (or a zero vector); negative: θ obtuse ($\frac{\pi}{2} < \theta \le \pi$).
Ex: $(\hat{i}+\hat{j})\cdot(\hat{i}-2\hat{j}) = -1 < 0$: obtuse angle.

Grade - Medium
Practice map, main concept: Sheet: J-M 137, 169 | DPP: 2(4); 3(5)
Also used in: none

Question 70
Does order of vectors matter in dot product?

Model Answer
No: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ (commutative). It is also distributive: $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 71
What is the formula of $\vec{a} \cdot \vec{b}$ if $\vec{a}$ and $\vec{b}$ are expressed in $\hat{i}, \hat{j}, \hat{k}$ form?

Model Answer

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

(From $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$, $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$.) Ex: $(2\hat{i} - \hat{j} + 3\hat{k}) \cdot (\hat{i} + 4\hat{j} - \hat{k}) = -5$.

Grade - Medium
Practice map, main concept: Sheet: J-II 10; C-I 15; C-II 33
Also used in: Sheet: J-I 16, 31; J-II 8, 22; J-M 1-2, 78, 137, 154, 169, 172; J-A 2; C-I 23; C-II 23 | DPP: 2(4); 3(5); 4(12); 5(11); 8(4, 15) | Question Bank: 20, 25, 33, 37, 64

Question 72
What is the condition for two vectors to be perpendicular?

Model Answer

$$\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0 \iff a_1 b_1 + a_2 b_2 + a_3 b_3 = 0$$

(For non-zero vectors.) Ex: $2\hat{i} + \lambda\hat{j} + \hat{k} \perp \hat{i} - 2\hat{j} + 3\hat{k}$: $5 - 2\lambda = 0 \Rightarrow \lambda = \frac{5}{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 9, 16 | DPP: 1(9); 3(7); 4(11) | Question Bank: 16
Also used in: Sheet: J-I 2; J-II 19, 59, 66; J-M 8, 10, 14-15, 17, 76, 82, 85, 116, 130, 157, 160; J-A 16, 21, 24, 32; C-I 1-2, 5, 12; C-II 7, 24, 41, 45 | DPP: 1(4); 2(2); 3(3); 4(2, 7, 10); 5(2, 6); 7(5); 8(4); 9(2) | Question Bank: 6, 68

Question 73
What do you mean by squaring a vector?

Model Answer

$$\vec{a}^2 = \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

It is the dot product of the vector with itself (a scalar). Ex: $(2\hat{i} - \hat{j} + 2\hat{k})^2 = 9$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 13

Question 74
What pattern is followed while we do squaring of sum / difference of two or more vectors?

Model Answer

$$|\vec{a} \pm \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \pm 2\,\vec{a}\cdot\vec{b}, \qquad (\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b}) = |\vec{a}|^2 - |\vec{b}|^2$$
$$|\vec{a}+\vec{b}+\vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$$

Same as algebra, with squares $\to |\ |^2$ and products $\to$ dot products. Ex: $\vec{a}, \vec{b}, \vec{c}$ mutually perpendicular unit vectors: $|2\vec{a} + 3\vec{b} + 6\vec{c}| = \sqrt{4+9+36} = 7$.

Grade - Medium

Practice map, main concept: Sheet: J-I 5; J-II 6-7; J-M 14, 131; J-A 18; C-II 34-35 | DPP: 3(3, 8, 10); 4(6); 8(16) | Question Bank: 7, 49

Also used in: Sheet: J-I 8, 34; J-II 5, 17; J-M 18, 85, 126, 128, 138, 160, 163, 166; J-A 1, 28, 34, 36, 40, 44; C-I 7, 16; C-II 8, 27, 29, 36, 40-41, 44-45 | DPP: 3(1, 4); 4(1, 3, 7); 5(3, 5) | Question Bank: 34-35, 39, 47-48, 59, 65

Question 75
How to find magnitude of a vector which cannot be expressed in $\hat{i}, \hat{j}, \hat{k}$ form?

Model Answer

$$|\vec{v}| = \sqrt{\vec{v}\cdot\vec{v}}$$

Square it, expand by the dot product pattern, use the given magnitudes and angles, then take the root.

Ex: $|\vec{a}| = 4$, $|\vec{b}| = 3$, angle $120°$: $|\vec{a}+\vec{b}|^2 = 16 + 9 - 12 = 13 \Rightarrow |\vec{a}+\vec{b}| = \sqrt{13}$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 6

Application of Dot product

Question 76
How to find angle between two vectors?

Model Answer

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2+a_2^2+a_3^2}\sqrt{b_1^2+b_2^2+b_3^2}}$$

Ex: $4\hat{i} + \hat{j} + 3\hat{k}$ and $2\hat{i} + 2\hat{j} - \hat{k}$: $\theta = \cos^{-1}\frac{7}{3\sqrt{26}}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 6-7; J-II 9; J-M 126-128, 138, 159-160; J-A 28; C-I 19; C-II 8, 28, 30, 40, 44 | DPP: 2(1-2); 3(1); 4(1, 9-10); 5(5-6); 9(2) | Question Bank: 32, 47-48, 65

Also used in: Sheet: J-I 9, 32-33; J-II 50; J-M 72, 141; J-A 41; C-I 10, 14, 16; C-II 15, 43 | DPP: 3(8); 4(6); 9(7) | Question Bank: 4, 7, 50

Question 77
How to find angle between two lines whose equations are known?

Model Answer

$$L_1 : \vec{r} = \vec{a} + \lambda\vec{p}, \quad L_2 : \vec{r} = \vec{b} + \mu\vec{q} : \qquad \cos\theta = \frac{|\vec{p} \cdot \vec{q}|}{|\vec{p}||\vec{q}|}$$

Only the direction vectors matter; the modulus gives the acute angle. Ex: $\vec{p} = (1, 2, 2)$, $\vec{q} = (2, -1, 2)$: $\cos\theta = \frac{4}{9}$.

Grade - Medium
Practice map, main concept: Sheet: J-II 26; C-I 22 | DPP: 3(11) | Question Bank: 6
Also used in: Sheet: J-A 22, 39 | DPP: 4(4) | Question Bank: 14

Question 78
What does projection of one vector along another vector represent?

Model Answer
The signed length of the shadow of $\vec{p}$ on the line of $\vec{q}$ (foot of perpendiculars from the ends of $\vec{p}$): $|\vec{p}|\cos\theta$. It is a scalar.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 11

Question 79
What is the formula for projection of $\vec{p}$ along $\vec{q}$?

Model Answer

$$\text{projection of } \vec{p} \text{ along } \vec{q} = |\vec{p}|\cos\theta = \frac{\vec{p} \cdot \vec{q}}{|\vec{q}|} = \vec{p} \cdot \hat{q}$$

Ex: $\vec{p} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ along $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$: $\frac{10}{\sqrt{6}}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 9, 16, 67, 136; J-A 34 | DPP: 4(7) | Question Bank: 1, 22, 51
Also used in: Sheet: J-I 51; J-M 4, 7, 167; J-A 41; C-I 13

Question 80
Can projection be negative? If yes, then explain why.

Model Answer
Yes. Projection $= |\vec{p}|\cos\theta$ is negative when θ is obtuse: the shadow falls opposite to $\vec{q}$. Its modulus is the length of the shadow.
Ex: projection of $\hat{i} - 2\hat{j}$ along $\hat{i} + \hat{j}$ is $-\frac{1}{\sqrt{2}}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 81

How to find vector component of one vector along / perpendicular to another vector?

Model Answer

$$\vec{p}_{\parallel} = (\vec{p}\cdot\hat{q})\,\hat{q} = \frac{\vec{p}\cdot\vec{q}}{|\vec{q}|^2}\,\vec{q}, \qquad \vec{p}_{\perp} = \vec{p} - \vec{p}_{\parallel}$$

Ex: $\vec{p} = 5\hat{i} - 2\hat{j} + 5\hat{k}$, $\vec{q} = 3\hat{i} + \hat{k}$: $\vec{p}_{\parallel} = 2\vec{q} = 6\hat{i} + 2\hat{k}$, $\vec{p}_{\perp} = -\hat{i} - 2\hat{j} + 3\hat{k}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 10, 51; J-II 11; J-M 2; J-A 38; C-I 5; C-II 29 | DPP: 2(5) | Question Bank: 4, 24, 58

Also used in: Sheet: J-M 16

Question 82

How to find foot of perpendicular from a point on a line whose equation is given?

Model Answer

Take the general point $Q = \vec{a} + \lambda\vec{b}$; put $\overrightarrow{PQ}\cdot\vec{b} = 0$, solve for λ; then Q is the foot.

Ex: $P(1,6,3)$, $\vec{r} = \hat{j} + 2\hat{k} + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$: $14\lambda - 14 = 0$, $\lambda = 1$, foot $(1,3,5)$.

Grade - Medium

Practice map, main concept: Sheet: J-II 36-37, 68; J-M 27, 31, 57, 83, 102, 106, 113, 146, 157, 194; J-A 16; C-II 4 | Question Bank: 8, 13, 15, 27, 31, 61, 68

Also used in: Sheet: J-II 28, 35; J-M 55, 112, 133, 150, 153; J-A 3; C-II 47 | Question Bank: 10

Question 83

How to find image of a point in a line whose equation is given?

Model Answer

$$\text{image } P' = 2Q - P \quad (Q = \text{foot of perpendicular, mid point of } PP')$$

Ex: $P(1,6,3)$, foot $Q(1,3,5)$ (above): image $P'(1,0,7)$.

Grade - Medium

Practice map, main concept: Sheet: J-M 144, 148, 150, 185, 187, 192, 196-197 | DPP: 3(9); 7(14) | Question Bank: 52

Also used in: none

Question 84

How to find distance of a point from a line by formula of cross product?

Model Answer

$$d = \frac{|\overrightarrow{AP}\times\vec{b}|}{|\vec{b}|} = \frac{|(\vec{p}-\vec{a})\times\vec{b}|}{|\vec{b}|}$$

$A(\vec{a})$ = point on the line $\vec{r} = \vec{a} + \lambda\vec{b}$ (since $d = |\overrightarrow{AP}|\sin\theta$).

Ex: $P(1,6,3)$, line above: $\overrightarrow{AP}\times\vec{b} = 13\hat{i} - 2\hat{j} - 3\hat{k}$, $d = \frac{\sqrt{182}}{\sqrt{14}} = \sqrt{13}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 29, 52, 54, 112, 133, 152, 182; C-II 9 | DPP: 3(2); 5(10); 8(8)

Also used in: Sheet: J-M 87, 153; J-A 22 | DPP: 5(7) | Question Bank: 23

Linear Combination theorem in plane

Question 85

Explain linear combination theorem in plane.

Model Answer

If $\vec{a}, \vec{b}$ are non-zero and non-collinear, every vector $\vec{r}$ coplanar with them can be written uniquely as $\vec{r} = x\vec{a} + y\vec{b}$.

Ex: $\hat{i} + 3\hat{j} + 2\hat{k} = (\hat{i} + \hat{j}) + 2(\hat{j} + \hat{k})$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 86

How to assume a vector in plane of two vectors $\vec{a}$ and $\vec{b}$?

Model Answer

$$\vec{r} = x\vec{a} + y\vec{b} \qquad (x, y \in \mathbb{R})$$

Ex: in the plane of $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$: $x\hat{i} + (x + y)\hat{j} + y\hat{k}$.

Grade - Low

Practice map, main concept: Sheet: J-II 17; C-I 13; C-II 7, 15, 41

Also used in: Sheet: J-I 22; C-II 30, 42

Question 87

How to calculate unknown scalars in questions of linear combination theorem in plane?

Model Answer

Write $\vec{r} = x\vec{a} + y\vec{b}$; take dot product with $\vec{a}$ and with $\vec{b}$ (or a suitable vector), or compare components, to get two equations; add any given condition (unit, perpendicular).

Ex: $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = \hat{i} + 2\hat{j}$; unit $\vec{c} = x\vec{a} + y\vec{b} \perp \vec{a}$: $2x - y = 0$, $|\vec{c}| = 1 \Rightarrow \vec{c} = \pm\frac{\hat{i}+\hat{j}}{\sqrt{2}}$.

Grade - Medium

Practice map, main concept: DPP: 4(2) | Question Bank: 19

Also used in: Sheet: J-M 68

Cross Product

Question 88

Write expression for cross product.

Model Answer

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin\theta\, \hat{n}$$

θ = angle between $\vec{a}, \vec{b}$ $(0 \le \theta \le \pi)$, $\hat{n}$ = unit vector perpendicular to both; the result is a vector, $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$.

Ex: $|\vec{a}| = 2$, $|\vec{b}| = 3$, $\theta = 30^\circ \Rightarrow |\vec{a} \times \vec{b}| = 3$.

Grade - Medium

Practice map, main concept: Sheet: J-I 14, 29, 34; J-M 22, 85, 132, 163, 175; J-A 15, 25, 36; C-I 20 | DPP: 5(2, 4, 7) | Question Bank: 34-35, 39, 50

Also used in: Sheet: J-I 7; J-M 60, 79, 84; J-A 18, 57; C-I 5, 23; C-II 8 | DPP: 8(5, 8, 10) | Question Bank: 20, 48, 55, 59, 65

Question 89
What is the direction of $\vec{a} \times \vec{b}$?

Model Answer
Perpendicular to the plane of $\vec{a}$ and $\vec{b}$, in the sense given by the right-hand rule: curl the fingers from $\vec{a}$ towards $\vec{b}$, the thumb points along $\vec{a} \times \vec{b}$.
So $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$: $(\vec{a} \times \vec{b}) \cdot \vec{a} = (\vec{a} \times \vec{b}) \cdot \vec{b} = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-M 13; J-A 21
Also used in: Sheet: J-M 131, 163; J-A 28, 52; C-I 19 | DPP: 8(1) | Question Bank: 5, 34, 49

Question 90
What is the relation between $\vec{a} \times \vec{b}$ and $\vec{b} \times \vec{a}$?

Model Answer

$$\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$$

Same magnitude, opposite direction: the cross product is not commutative (it is distributive over addition).

Grade - Medium
Practice map, main concept: Sheet: J-I 15, 28; J-M 79 | DPP: 4(5)
Also used in: Sheet: J-II 18, 62 | DPP: 5(4)

Question 91
How to compute $\vec{a} \times \vec{b}$ when $\vec{a}$ and $\vec{b}$ are expressed in $\hat{i}, \hat{j}, \hat{k}$ form?

Model Answer

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\hat{i} - (a_1b_3 - a_3b_1)\hat{j} + (a_1b_2 - a_2b_1)\hat{k}$$

Ex: $(\hat{i} + 2\hat{j} + 3\hat{k}) \times (2\hat{i} + \hat{j} - \hat{k}) = -5\hat{i} + 7\hat{j} - 3\hat{k}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 40; J-A 29
Also used in: Sheet: J-I 34, 40, 47; J-II 15-16, 27, 53, 66; J-M 9, 11, 24, 70, 74, 77, 139, 166, 173, 176-177; J-A 34, 44, 52; C-I 8; C-II 17 | DPP: 8(11) | Question Bank: 22, 35, 38, 40, 66

Question 92
What is the relation between $\vec{a}$ and $\vec{b}$ if $\vec{a} \times \vec{b} = \vec{0}$?

Model Answer
Either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, or $\vec{a} \parallel \vec{b}$ ($\theta = 0$ or π), i.e. $\vec{b} = \lambda\vec{a}$. In particular $\vec{a} \times \vec{a} = \vec{0}$.

Grade - Medium
Practice map, main concept: Sheet: J-II 18, 62; J-A 2 | DPP: 7(11) | Question Bank: 25, 55
Also used in: Sheet: J-I 30; J-II 17, 60-61; J-M 1, 18, 20, 74, 164, 171, 178, 180 | DPP: 4(5)

Question 93
What is cross product of two vectors taken from $\hat{i}, \hat{j}, \hat{k}$?

Model Answer

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}, \quad \hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}, \quad \hat{k} \times \hat{i} = \hat{j}$$

Reverse (anticyclic) order gives the negative: $\hat{j} \times \hat{i} = -\hat{k}$, $\hat{k} \times \hat{j} = -\hat{i}$, $\hat{i} \times \hat{k} = -\hat{j}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 94
How to assume a vector perpendicular to $\vec{a}$ and $\vec{b}$?

Model Answer

$$\vec{r} = \lambda(\vec{a} \times \vec{b}), \qquad \text{unit: } \pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

Valid when $\vec{a}, \vec{b}$ are non-collinear. Ex: $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$: $\vec{a} \times \vec{b} = \hat{i} - \hat{j} + \hat{k}$, unit vectors $\pm\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$.

Grade - Medium
Practice map, main concept: Sheet: J-II 15, 29, 53, 58; J-M 60, 70, 130; J-A 9; C-II 23 | DPP: 4(12); 5(8); 8(15); 10(12) | Question Bank: 30
Also used in: Sheet: J-I 13; J-M 67, 83, 133; J-A 11, 27; C-I 17 | DPP: 10(3)

Question 95
If $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$, then find $\vec{a}$ in terms of $\vec{b}$ and $\vec{c}$.

Model Answer

$$\vec{a} \times (\vec{b} - \vec{c}) = \vec{0} \Rightarrow \vec{a} = \lambda(\vec{b} - \vec{c}) \quad (\vec{a} \neq \vec{0},\ \vec{b} \neq \vec{c})$$

We cannot cancel $\vec{a}$ to get $\vec{b} = \vec{c}$; we only get $\vec{a} \parallel (\vec{b} - \vec{c})$, i.e. $\vec{b} - \vec{c} = \mu\vec{a}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 17; J-II 19; J-M 8, 15, 17, 76, 82, 134, 141; C-I 2 | Question Bank: 64
Also used in: DPP: 5(1)

Application of cross product

Question 96
How to express equation of line, expressed in cross product form, into vector form?

Model Answer

$$\vec{r} \times \vec{b} = \vec{a} \times \vec{b} \Rightarrow (\vec{r} - \vec{a}) \times \vec{b} = \vec{0} \Rightarrow \vec{r} = \vec{a} + \lambda\vec{b}$$

If given as $\vec{r} \times \vec{b} = \vec{c}$ (with $\vec{b} \cdot \vec{c} = 0$), a point on it is $\dfrac{\vec{b} \times \vec{c}}{|\vec{b}|^2}$.

Ex: $\vec{r} \times (\hat{i} + \hat{j} + \hat{k}) = -\hat{i} + 2\hat{j} - \hat{k}$: point $-\hat{i} + \hat{k}$, so $\vec{r} = -\hat{i} + \hat{k} + \lambda(\hat{i} + \hat{j} + \hat{k})$.

Grade - Low
Practice map, main concept: Sheet: J-I 12
Also used in: none

Question 97
How to find area of triangle if side vectors are known?

Model Answer

$$\Delta = \tfrac{1}{2}|\vec{a} \times \vec{b}|$$

$\vec{a}, \vec{b}$ = any two side vectors of the triangle. Ex: $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$: $\Delta = \tfrac{1}{2}|\hat{i} - \hat{j} + \hat{k}| = \tfrac{\sqrt{3}}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 69; J-M 78, 173, 177; J-A 41; C-II 32 | DPP: 8(13) | Question Bank: 9, 14, 38, 62

Also used in: Sheet: J-II 14; J-M 52; J-A 12, 54 | DPP: 5(9)

Question 98
How to find area of triangle if vertices are given?

Model Answer

$$\Delta ABC = \tfrac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$$

Ex: $A(1,1,1), B(1,2,3), C(2,3,1)$: $\overrightarrow{AB} \times \overrightarrow{AC} = -4\hat{i} + 2\hat{j} - \hat{k}$, area $\tfrac{\sqrt{21}}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 12, 16, 31, 50; J-M 55; C-II 11 | Question Bank: 17, 57

Also used in: Sheet: J-II 33, 40, 57, 68; J-M 110, 187; J-A 35; C-I 21 | DPP: 1(10); 8(13) | Question Bank: 15, 18

Question 99
Write expression of area of triangle whose vertices are $\vec{a}, \vec{b}$ and $\vec{c}$.

Model Answer

$$\Delta = \tfrac{1}{2}|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$$

It comes from $\tfrac{1}{2}|(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})|$; the points are collinear iff $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 13

Also used in: Sheet: J-A 41

Question 100
How to find area of a parallelogram?

Model Answer

$$\text{Area} = |\vec{a} \times \vec{b}| = \tfrac{1}{2}|\vec{d_1} \times \vec{d_2}|$$

$\vec{a}, \vec{b}$ = adjacent side vectors, $\vec{d_1}, \vec{d_2}$ = diagonal vectors. Ex: sides $\hat{i} + \hat{j}$, $\hat{j} + \hat{k}$: area $\sqrt{3}$ (diagonals $\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} - \hat{k}$ give the same).

Grade - Medium

Practice map, main concept: Sheet: J-M 129, 176 | Question Bank: 40

Also used in: Sheet: J-II 13; J-M 125, 174; J-A 12, 38; C-II 21 | Question Bank: 58

Question 101
How to find area of quadrilateral?

Model Answer

$$\text{Area of } ABCD = \tfrac{1}{2}|\overrightarrow{AC} \times \overrightarrow{BD}|$$

Half the magnitude of the cross product of the diagonal vectors (plane quadrilateral); or split it into two triangles by a diagonal and add.

Grade - Medium
Practice map, main concept: Sheet: J-II 13; J-M 56, 91, 125, 162, 165, 174 | DPP: 8(11)
Also used in: none

Direction Cosines and Direction ratios

Question 102
Define direction angles and direction cosines.

Model Answer
Direction angles α, β, γ: angles made by a directed line with the positive x-, y-, z-axes ($0 \le \alpha, \beta, \gamma \le \pi$). Direction cosines: $\cos\alpha, \cos\beta, \cos\gamma$.

Reversing the direction changes all the direction cosines in sign.

Grade - Medium
Practice map, main concept: Sheet: J-I 36; J-M 12 | Question Bank: 12
Also used in: Sheet: J-M 168; C-I 18 | Question Bank: 36, 52

Question 103
How are direction cosines generally denoted?

Model Answer

$$l = \cos\alpha,\ m = \cos\beta,\ n = \cos\gamma, \qquad l^2 + m^2 + n^2 = 1$$

Hence also $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$ and $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 104
How to find direction cosines of any direction if vector along that direction is known and vice versa?

Model Answer

$$\vec{r} = a\hat{i} + b\hat{j} + c\hat{k} \ \Rightarrow\ l, m, n = \frac{a}{|\vec{r}|}, \frac{b}{|\vec{r}|}, \frac{c}{|\vec{r}|}$$

Conversely, a vector of magnitude k along DCs l, m, n is $k(l\hat{i} + m\hat{j} + n\hat{k})$. Ex: $2\hat{i} - \hat{j} - 2\hat{k}$ ($|\vec{r}| = 3$): DCs $\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}$.

Grade - Medium
Practice map, main concept: Sheet: J-II 5, 57 | DPP: 6(14)
Also used in: Sheet: J-I 41; J-M 36, 144, 158 | DPP: 7(12) | Question Bank: 12

Question 105
What are direction cosines of coordinate axes?

Model Answer

$$x\text{-axis} : 1, 0, 0; \quad y\text{-axis} : 0, 1, 0; \quad z\text{-axis} : 0, 0, 1$$

(Negative axes: $-1, 0, 0$ etc.)

Grade - Low
Practice map, main concept: none
Also used in: none

Question 106
Define direction ratios.

Model Answer
Any three numbers a, b, c proportional to the direction cosines: $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$. They are not unique (any non-zero multiple works).

$$l = \pm\frac{a}{\sqrt{a^2+b^2+c^2}},\ m = \pm\frac{b}{\sqrt{a^2+b^2+c^2}},\ n = \pm\frac{c}{\sqrt{a^2+b^2+c^2}}$$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 107
How to find direction ratios of any direction if vector along that direction is known and vice versa?

Model Answer
DRs of $a\hat{i}+b\hat{j}+c\hat{k}$ are a, b, c (its components); conversely a vector with DRs a, b, c is $\lambda(a\hat{i}+b\hat{j}+c\hat{k})$.
DRs of AB: $x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1$.
Ex: $A(1, 2, 3), B(3, 1, 1)$: DRs of AB are $2, -1, -2$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 108
What is the relation between direction cosines and unit vector along that direction?

Model Answer

$$\hat{r} = l\hat{i} + m\hat{j} + n\hat{k}$$

The direction cosines are the components of the unit vector along that direction.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 109

How to find direction cosines of a line if two relations involving l, m and n are given?

Model Answer

Eliminate one of them (say n) using the linear relation, factorise the resulting quadratic in l, m, get two sets of ratios $l : m : n$, then normalise using $l^2 + m^2 + n^2 = 1$.

Ex: $l + m + n = 0,\ l^2 + m^2 - n^2 = 0 \Rightarrow lm = 0$: DRs $0, 1, -1$ and $1, 0, -1$; DCs $0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ and $\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}$.

Grade - Low

Practice map, main concept: Sheet: J-I 37; J-II 23

Also used in: none

Question 110

How to find angle between two lines if direction ratios of lines are known?

Model Answer

$$\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Perpendicular: $a_1a_2 + b_1b_2 + c_1c_2 = 0$; parallel: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$. Ex: DRs $1, 1, 0$ and $0, 1, 1$: $\cos\theta = \frac{1}{2}$, $\theta = 60°$.

Grade - Medium

Practice map, main concept: Sheet: J-I 38; J-M 183

Also used in: Sheet: J-I 37; J-II 23; J-A 30

Symmetric Form of Line

Question 111

How to write symmetric form of line and what to keep in mind?

Model Answer

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Line through (x_1, y_1, z_1) with DRs a, b, c. Keep the coefficients of x, y, z equal to $+1$ before reading the point and DRs; a zero denominator only means that numerator is 0 (e.g. $c = 0 \Rightarrow z = z_1$); if a, b, c are DCs, the common value is the distance from (x_1, y_1, z_1).

Ex: $\frac{2x-1}{3} = \frac{1-y}{2} = z$ is $\frac{x-\frac{1}{2}}{3/2} = \frac{y-1}{-2} = \frac{z}{1}$: point $(\frac{1}{2}, 1, 0)$, DRs $3, -4, 2$.

Grade - Medium

Practice map, main concept: DPP: 7(2, 5)

Also used in: Sheet: J-II 29, 58; J-M 32, 37 | Question Bank: 6, 28, 52

Question 112

How to convert symmetric form of line to vector form?

Model Answer

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = \lambda \ \Rightarrow\ \vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k})$$

Ex: $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-1}$: $\vec{r} = \hat{i} - 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 113
What to keep in mind while converting several different symmetric forms to vector forms?

Model Answer
Use a different parameter for each line $(\lambda, \mu, \dots)$: the lines are independent, so the same parameter value does not give corresponding points. Also make each line's coefficients of x, y, z unity first.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 155, 182-183, 189

Shortest Distance between two lines

Question 114
What does the shortest distance between two lines mean?

Model Answer
The least value of PQ for P on one line and Q on the other; it is the length of the common perpendicular segment of the two lines.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 115
What is the shortest distance between two intersecting lines?

Model Answer
Zero, since they have a common point. Conversely, $SD = 0$ for non-parallel lines means they intersect.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 116
How to find shortest distance between two parallel lines?

Model Answer

$$\vec{r} = \vec{a}_1 + \lambda\vec{b},\ \vec{r} = \vec{a}_2 + \mu\vec{b}: \quad d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$$

It is the distance of a point of one line from the other line. Ex: $\vec{r} = \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$, $\vec{r} = 3\hat{i} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$: $d = \frac{|-6\hat{j}+6\hat{k}|}{3} = 2\sqrt{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 188
Also used in: none

Question 117
What is the formula of shortest distance between two skew lines?

Model Answer

$$\vec{r} = \vec{a}_1 + \lambda\vec{b}_1,\ \vec{r} = \vec{a}_2 + \mu\vec{b}_2 : \quad SD = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

Ex: $\vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} - \hat{j} + \hat{k})$, $\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$: $\vec{b}_1 \times \vec{b}_2 = 3\hat{i} - \hat{j} - 7\hat{k}$, $SD = \frac{10}{\sqrt{59}}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 54, 64; J-M 32, 35, 37, 39, 45, 47, 49-50, 61, 88, 93, 109, 143, 147, 149, 155, 170, 186, 190, 195 | Question Bank: 3, 10

Also used in: Sheet: J-II 39, 68; J-A 47; C-I 20; C-II 9, 12 | DPP: 10(10)

Question 118
What does formula of shortest distance between two skew lines represent?

Model Answer
The length of the projection of $\overrightarrow{AB} = \vec{a}_2 - \vec{a}_1$ (joining a point of each line) on the common perpendicular direction $\vec{b}_1 \times \vec{b}_2$.

Hence the lines intersect (are coplanar) iff $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 119
How to find equation of line of shortest distance between two skew lines?

Model Answer
Take general points $P(\lambda)$ on L_1 and $Q(\mu)$ on L_2; solve $\overrightarrow{PQ} \cdot \vec{b}_1 = 0$, $\overrightarrow{PQ} \cdot \vec{b}_2 = 0$ for λ, μ; the line PQ is the line of SD and PQ = SD.

Ex: $\vec{r} = \lambda(\hat{i} + \hat{j})$, $\vec{r} = \hat{i} + 2\hat{k} + \mu(\hat{j} + \hat{k})$: $\lambda = 0,\ \mu = -1,\ P(0, 0, 0),\ Q(1, -1, 1)$; line $\vec{r} = t(\hat{i} - \hat{j} + \hat{k})$, SD $= \sqrt{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 140, 154, 193; J-A 32 | Question Bank: 26

Also used in: Sheet: J-M 77

Plane

Question 120
Define plane.

Model Answer
A surface such that the line segment joining any two of its points lies completely on it.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 121
What is normal vector of a plane?

Model Answer
A non-zero vector $\vec{n}$ perpendicular to the plane, i.e. perpendicular to every line lying in the plane; any non-zero multiple of $\vec{n}$ is also a normal.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 122
What is general equation of plane?

Model Answer

$$ax + by + cz + d = 0 \;\; (a, b, c \text{ not all } 0), \qquad \vec{r} \cdot \vec{n} = q$$

Every first degree equation in x, y, z represents a plane. Through (x_1, y_1, z_1): $a(x-x_1)+b(y-y_1)+c(z-z_1) = 0$, i.e. $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 65; J-M 40; J-A 11, 43, 49, 51

Question 123
What is the relation between normal vector and cartesian equation of plane?

Model Answer
The coefficients of x, y, z are the components (DRs) of a normal: $ax + by + cz + d = 0$ has normal $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$.
Ex: $3x + 2y + 5z - 6 = 0$ has normal $3\hat{i} + 2\hat{j} + 5\hat{k}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 43, 58
Also used in: Sheet: J-II 52; J-M 12, 41, 53 | DPP: 6(1); 7(15)

Question 124
How to convert dot product form of equation of plane to cartesian form?

Model Answer
Put $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ in $\vec{r} \cdot \vec{n} = q$ (or $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$) and expand the dot product.
Ex: $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 3 \Rightarrow 2x - y + z = 3$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 6(2)

Question 125
What to keep in mind while finding cartesian equation of plane?

Model Answer
Find two things: one point (x_1, y_1, z_1) on the plane and a normal (a, b, c); then write $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$. The normal is needed only up to a multiple, so simplify it first.
If the normal is unknown, take it as the cross product of two non-parallel vectors lying in (parallel to) the plane.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 126
How to find normal vector of a plane if two vectors parallel to the plane are given?

Model Answer

$$\vec{n} = \vec{p} \times \vec{q}$$

$\vec{p}, \vec{q}$ non-collinear. Ex: $\vec{p} = \hat{i} + \hat{j}$, $\vec{q} = \hat{j} + \hat{k}$: $\vec{n} = \hat{i} - \hat{j} + \hat{k}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 40; J-II 34, 39, 41, 52; J-M 25, 36, 38, 99, 105; J-A 22, 47 | DPP: 6(1, 4-5, 8); 7(8)

Also used in: Sheet: J-I 47; J-II 30-31, 35-37, 43, 55, 57; J-M 72, 90, 97, 101, 108, 113; J-A 5, 23, 53; C-I 10 | DPP: 6(10); 9(4)

Question 127
Write equations of coordinate planes.

Model Answer

$$yz\text{-plane} : x = 0, \quad zx\text{-plane} : y = 0, \quad xy\text{-plane} : z = 0$$

Planes parallel to them: $x = k$, $y = k$, $z = k$.

Grade - Low

Practice map, main concept: DPP: 8(14)

Also used in: none

Question 128
How to find point of intersection of a line and a plane?

Model Answer
Put the general point of the line in the equation of the plane and solve for the parameter. No solution: line $\parallel$ plane; identity: line lies in the plane.
Ex: $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} = t$ and $x - y + z = 5$: $11t = 0$, point $(2, -1, 2)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 41; J-II 32; J-M 94, 119; J-A 3, 35, 50 | DPP: 6(16); 7(7, 10)

Also used in: Sheet: J-I 45, 50; J-M 26, 55, 100, 115; J-A 54

Angle between two planes

Question 129
How to find angle between two planes?

Model Answer

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|} = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

It is the angle between the normals. Ex: $x + y + 2z = 9$, $2x - y + z = 15$: $\cos\theta = \frac{3}{6} = \frac{1}{2}$, $\theta = 60°$.

Grade - Medium

Practice map, main concept: Sheet: J-M 19, 23 | DPP: 9(4); 10(2)

Also used in: Sheet: J-II 36; J-A 30

Question 130
How to check whether two given planes are parallel?

Model Answer

$$\vec{n}_1 \parallel \vec{n}_2 : \quad \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

If also $= \frac{d_1}{d_2}$, the planes coincide. Ex: $2x - y + 3z = 1$ and $4x - 2y + 6z = 7$ are parallel.

Grade - Low
Practice map, main concept: DPP: 6(9)
Also used in: Sheet: J-I 46

Question 131
How to check whether two given planes are perpendicular?

Model Answer

$$\vec{n}_1 \cdot \vec{n}_2 = 0 : \quad a_1a_2 + b_1b_2 + c_1c_2 = 0$$

Ex: $x + y + z = 1$ and $x - 2y + z = 3$: $1 - 2 + 1 = 0$, perpendicular.

Grade - Medium
Practice map, main concept: Sheet: J-A 27
Also used in: Sheet: J-I 40; J-II 34, 44; J-M 46, 89, 99, 103

Angle between line and plane

Question 132
How to find angle between line and plane?

Model Answer

$$\sin\theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$$

θ is the complement of the angle between the line and the normal. Ex: DRs $2, 3, 6$ and plane $10x + 2y - 11z = 3$: $\sin\theta = \frac{40}{7 \cdot 15} = \frac{8}{21}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 92, 115; C-I 21 | DPP: 6(6, 15)
Also used in: Sheet: J-M 23

Question 133
How to check whether a line is perpendicular to a plane?

Model Answer

$$\vec{b} \parallel \vec{n} : \quad \frac{l}{a} = \frac{m}{b} = \frac{n}{c}$$

The line's direction (l, m, n) must be proportional to the normal (a, b, c), i.e. $\vec{b} \times \vec{n} = \vec{0}$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 31; J-M 37

Question 134
How to check whether a line is parallel to a plane?

Model Answer

$$\vec{b} \cdot \vec{n} = 0 : \quad al + bm + cn = 0$$

The line is perpendicular to the normal; if in addition a point of the line is not on the plane, it is strictly parallel (otherwise it lies in the plane).

Grade - Medium
Practice map, main concept: Sheet: J-I 43; J-II 25, 51; J-M 42 | DPP: 7(9)
Also used in: Sheet: J-II 38; J-M 38, 86-87, 118 | DPP: 7(3); 10(12)

Question 135
How to check whether a line is contained in a plane?

Model Answer
Two conditions: (i) $\vec{b} \cdot \vec{n} = 0$, (ii) one point of the line satisfies the plane.
Ex: line through $(1, 1, 1)$ with DRs $1, -2, 1$ and plane $x + y + z = 3$: $1 - 2 + 1 = 0$ and $1 + 1 + 1 = 3$, so it lies in the plane.

Grade - Medium
Practice map, main concept: Sheet: J-M 51 | DPP: 6(13); 7(3)
Also used in: Sheet: J-A 45

Equation of plane

Question 136
Write expression for parametric form of equation of plane.

Model Answer

$$\vec{r} = \vec{a} + \lambda\vec{p} + \mu\vec{q}$$

Plane through the point $\vec{a}$ parallel to $\vec{p}$ and $\vec{q}$; λ, μ are real parameters.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 137
What do $\vec{p}$ and $\vec{q}$ represent in parametric form?

Model Answer
Two non-collinear vectors parallel to (lying in) the plane; the normal is $\vec{n} = \vec{p} \times \vec{q}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 138
How to convert parametric form of equation of plane into cartesian form?

Model Answer

$$\vec{n} = \vec{p} \times \vec{q}, \qquad (\vec{r} - \vec{a}) \cdot \vec{n} = 0$$

Ex: $\vec{r} = (1, 2, 1) + \lambda(2, 1, 3) + \mu(1, -1, 1)$: $\vec{n} = (4, 1, -3)$, plane $4x + y - 3z = 3$.

Grade - Medium
Practice map, main concept: DPP: 6(10, 12)
Also used in: Sheet: J-II 40; J-M 89; J-A 46

Question 139
Define x-intercept, y-intercept and z-intercept of a plane.

Model Answer
The x-intercept is the x-coordinate of the point where the plane meets the x-axis (put $y = z = 0$); similarly for y and z. Intercepts are signed numbers.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 140
What is intercept form of equation of plane and when to use it?

Model Answer

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

The plane meets the axes at $(a, 0, 0), (0, b, 0), (0, 0, c)$. Use it when intercepts or points on the axes are involved (e.g. triangle or tetrahedron cut off by the plane); not for planes through the origin or parallel to an axis.

Ex: intercepts $2, 3, 4$: $6x + 4y + 3z = 12$.

Grade - Low
Practice map, main concept: Sheet: J-I 42; C-I 11; C-II 5
Also used in: Sheet: J-II 33; C-II 3 | DPP: 8(12)

Question 141
How to find intercepts of plane on coordinate axes if cartesian form is given?

Model Answer
Write it as $ax + by + cz = k$ and divide by k: $\frac{x}{k/a} + \frac{y}{k/b} + \frac{z}{k/c} = 1$, so the intercepts are $\frac{k}{a}, \frac{k}{b}, \frac{k}{c}$ (or put two variables 0).

Ex: $2x + 3y - 4z = 12$: intercepts $6, 4, -3$.

Grade - Medium
Practice map, main concept: Sheet: J-M 41; C-II 1 | DPP: 6(2)
Also used in: Sheet: J-II 31, 41; J-M 42, 116; C-I 21

Distance of a Point from a plane

Question 142

What is the formula to find distance of a point from plane?

Model Answer

$$d = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}, \qquad d = \frac{|\vec{p} \cdot \vec{n} - q|}{|\vec{n}|} \text{ for } \vec{r} \cdot \vec{n} = q$$

Ex: $(1, 2, 3)$ from $2x - y + 2z + 3 = 0$: $d = \frac{|2-2+6+3|}{3} = 3$.

Grade - Medium

Practice map, main concept: Sheet: J-I 47; J-II 55; J-M 101; J-A 43; C-II 3 | DPP: 7(15)

Also used in: Sheet: J-I 42, 44, 48; J-II 44-45, 68; J-M 19, 25, 28, 42-44, 46, 48, 51, 62, 92, 94, 96, 103-104, 106, 112; J-A 4, 19, 22, 30, 47; C-I 11; C-II 1, 5, 9, 47 | DPP: 6(4, 12); 7(8)

Question 143

What is the formula to find distance between two parallel planes?

Model Answer

$$ax + by + cz + d_1 = 0,\ ax + by + cz + d_2 = 0: \quad d = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

First make the coefficients of x, y, z identical. Ex: $x + 2y - 2z + 1 = 0$, $2x + 4y - 4z + 5 = 0$ $(x + 2y - 2z + \frac{5}{2} = 0)$: $d = \frac{3/2}{3} = \frac{1}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 43 | DPP: 6(3)

Also used in: Sheet: J-A 49

Position of two points wrt a plane

Question 144

How to find relative positions of two given points with respect to a given plane?

Model Answer

Put both points in $P(x, y, z) = ax + by + cz + d$: same sign $\Rightarrow$ same side; opposite signs $\Rightarrow$ opposite sides (the plane divides the join internally); zero $\Rightarrow$ the point is on the plane.

Ex: $x + y + z - 1$ at $(1, 1, 1)$ and $(-1, 0, 0)$: 2 and -2, opposite sides.

Grade - Low

Practice map, main concept: none

Also used in: none

Angle Bisector Plane between two planes

Question 145

What is the definition of angle bisector plane of two planes?

Model Answer

The locus of points equidistant from two intersecting planes; it bisects the angle between them. There are two such planes, perpendicular to each other, both through the line of intersection.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 146
What is formula for angle bisector plane of two planes?

Model Answer

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}}=\pm\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}$$

Ex: $2x-y+2z+3=0$, $3x-2y+6z+8=0$: bisectors $5x-y-4z-3=0$ ('+') and $23x-13y+32z+45=0$ ('−').

Grade - Low
Practice map, main concept: none
Also used in: none

Question 147
How to differentiate acute and obtuse angle bisector from the equations of angle bisectors?

Model Answer
Make d_1, d_2 of the same sign (say positive). If $a_1a_2+b_1b_2+c_1c_2<0$, the '+' bisector (origin side) is the acute one; if >0, the '+' bisector is the obtuse one.

Or: if a bisector makes angle θ with one plane, it is the acute bisector when $\theta<45°$. Ex: above, $6+2+12=20>0$: $5x-y-4z-3=0$ is obtuse, $23x-13y+32z+45=0$ is acute.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 148
What do you mean by angle bisector containing a fixed point?

Model Answer
The two planes form two pairs of vertically opposite angular regions; it is the bisector of that angle (region between the planes) in which the given point lies, not a bisector plane passing through the point.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 149
How to find equation of angle bisector containing a fixed point?

Model Answer
Write the planes so that $P_1(x_1,y_1,z_1)$ and $P_2(x_1,y_1,z_1)$ have the same sign (multiply one by -1 if needed); then take the '+' sign: $\dfrac{P_1}{|\vec{n}_1|}=+\dfrac{P_2}{|\vec{n}_2|}$.

Ex: planes of 146, point $(-2,0,0)$: $P_1=-1$, $P_2=2$ (opposite), so the bisector is $23x-13y+32z+45=0$; for the origin $(3,8$ same sign) it is $5x-y-4z-3=0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Family of planes

Question 150

How to find the equation of a plane passing through the line of intersection of two planes $P_1 = 0$ and $P_2 = 0$?

Model Answer

$$P_1 + \lambda P_2 = 0$$

Find λ from the extra condition (a point, parallel / perpendicular to something, a distance).

Ex: through $(1, 1, 1)$, for $x + y + z = 1$, $2x + 3y + 4z = 5$: $\lambda = -\frac{1}{2}$, plane $y + 2z = 3$.

Grade - Medium

Practice map, main concept: Sheet: J-I 46, 48; J-II 38, 44-45; J-M 28, 46, 62, 86, 89, 103-104; J-A 4, 19

Also used in: DPP: 7(4)

Question 151

How to find the equation of a plane containing a line whose equation is given in unsymmetrical form?

Model Answer

Line $P_1 = 0 = P_2$ lies in every plane $P_1 + \lambda P_2 = 0$; find λ from the extra condition. (A symmetric-form line can first be written as two planes.)

Ex: through $(1, 1, 1)$ containing $x+y+z-6 = 0 = 2x+3y+4z+5$: $\lambda = \frac{3}{14}$, plane $20x+23y+26z-69 = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-M 48

Also used in: Sheet: J-M 102, 118

Line of intersection of two planes

Question 152

How to find the equation of the line of intersection of two given planes?

Model Answer

Direction $\vec{n}_1 \times \vec{n}_2$; a point: put $z = 0$ (or any convenient value) and solve the two equations. (Or put $z = \lambda$ and solve for x, y in terms of λ.)

Ex: $x + y + z = 1$, $2x + 3y + 4z = 5$: direction $(1, -2, 1)$, point $(-2, 3, 0)$: $\frac{x+2}{1} = \frac{y-3}{-2} = \frac{z}{1}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 27-28, 35, 42, 65, 70; J-M 53, 72; J-A 30, 45, 54; C-I 10; C-II 47 | DPP: 7(12)

Also used in: Sheet: J-II 32, 38, 47; J-M 27, 48, 54, 113; J-A 20 | DPP: 7(9)

Unsymmetrical Form of a line

Question 153
How is the unsymmetrical form of a line represented?

Model Answer

$$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$$

i.e. $P_1 = 0 = P_2$, where the normals (a_1, b_1, c_1) and (a_2, b_2, c_2) are not parallel.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 38; J-II 65; J-M 119 | Question Bank: 68

Question 154
What does the unsymmetrical form of a line represent?

Model Answer

It represents the straight line which is the line of intersection of the two (non-parallel) planes $P_1 = 0$ and $P_2 = 0$; its direction is $\vec{n}_1 \times \vec{n}_2$.

Ex: $x = 0 = y$ is the z-axis.

Grade - Low

Practice map, main concept: none

Also used in: none

Equation of plane

Question 155
How to find the distance of a point from a given plane measured parallel to a given line?

Model Answer

Through P draw the line parallel to the given line, $P + t\vec{b}$; put it in the plane to get t and the point Q; distance $= PQ = |t||\vec{b}|$.

Ex: $(1, -2, 3)$ from $x - y + z = 5$ parallel to $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$: $Q = \left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$, distance 1.

Grade - Medium

Practice map, main concept: Sheet: J-I 45; J-M 26, 77, 100, 118

Also used in: Sheet: C-II 2

Question 156
How to find the distance of a point from a given line measured parallel to a given plane?

Model Answer

Take the general point $Q(\lambda)$ of the line; $\overrightarrow{AQ}$ is parallel to the plane, so $\overrightarrow{AQ} \cdot \vec{n} = 0$ gives λ; distance $= |\overrightarrow{AQ}|$.

Ex: $A(1, -2, 3)$, line $\vec{r} = (0, 1, 0) + \lambda(2, 3, -6)$, plane $x - y + z = 5$: $\lambda = -1$, $\overrightarrow{AQ} = (-3, 0, 3)$, distance $3\sqrt{2}$.

Grade - Low

Practice map, main concept: Sheet: C-II 2

Also used in: none

Question 157
How to find the foot of perpendicular from a point on a given plane?

Model Answer

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=-\frac{ax_1+by_1+cz_1+d}{a^2+b^2+c^2}$$

(Line through the point along $\vec{n}$, meet the plane.) Ex: from $(1,3,4)$ on $2x-y+z+3=0$: foot $(-1,4,3)$.

Grade - Medium
Practice map, main concept: Sheet: J-I 44; J-II 40; J-M 108, 114; J-A 8, 20 | DPP: 10(7)
Also used in: Sheet: J-A 54; C-I 14

Concept of Image in 3D

Question 158
How to find the image of a point with respect to a given plane?

Model Answer

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=-\frac{2(ax_1+by_1+cz_1+d)}{a^2+b^2+c^2}$$

The foot N is the mid-point of AA'. Ex: $(1,3,4)$ in $2x-y+z+3=0$: image $(-3,5,2)$.

Grade - Medium
Practice map, main concept: Sheet: J-I 49; J-M 44, 90, 95, 97-98, 110, 117; J-A 23, 31, 37, 46 | DPP: 6(17); 7(13)
Also used in: Sheet: J-II 46 | DPP: 7(16)

Question 159
How to find the image of a line with respect to another parallel line?

Model Answer
The image is parallel to the given line: take a point A of the line, its image A' in the mirror line; image line $\vec{r}=\vec{a}\,'+t\vec{p}$.
Ex: $\vec{r}=2\hat{i}+t\hat{k}$ in the z-axis: $\vec{r}=-2\hat{i}+t\hat{k}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 160
How to find the image of a line with respect to another intersecting line?

Model Answer
Find the meeting point M; take another point A of the line and its image A' in the mirror line; image line $\vec{r}=\vec{a}\,'+t\,\overrightarrow{A'M}$.
Ex: x-axis in $\vec{r}=s(\hat{i}+\hat{j})$: $M=O$, image of $(1,0,0)$ is $(0,1,0)$, so the image is the y-axis.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 161
How to find the image of a line with respect to a plane if the line intersects the plane?

Model Answer
Find the meeting point M of line and plane; take another point A of the line and its image A' in the plane; the image line is MA'.

Ex: line through $(1,3,4)$ along $(1,1,0)$, plane $2x-y+z+3=0$: $M(-5,-3,4)$, $A'(-3,5,2)$, image $\frac{x+5}{1}=\frac{y+3}{4}=\frac{z-4}{-1}$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 162
How to find the image of a line with respect to a plane if the line is parallel to the plane?

Model Answer
The image is parallel to the line: image A' of one point A of the line in the plane, then $\vec{r}=\vec{a}\,'+t\vec{p}$.
Ex: line through $(1,3,4)$ along $(1,1,-1)$, plane $2x-y+z+3=0$: image $\frac{x+3}{1}=\frac{y-5}{1}=\frac{z-2}{-1}$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 46 | DPP: 7(16)
Also used in: none

Question 163
How to find the image of a plane with respect to another parallel plane?

Model Answer
Object $ax+by+cz=d_1$, mirror $ax+by+cz=d$: image $ax+by+cz=\delta$ with d_1,d,δ in A.P., i.e. $\delta=2d-d_1$ (make the coefficients equal first).

Ex: $x+2y+2z=1$ in $x+2y+2z=4$: $x+2y+2z=7$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 164
How to find the image of a plane with respect to another intersecting plane?

Model Answer
Image plane $P_1+\lambda P_m=0$ (it passes through the common line). Find λ: take a point of P_1, its image in P_m lies on the image plane (or: angle between P_1,P_m = angle between image and P_m).

Ex: $x+z=2$ in $x+y=1$: $(2,0,0)\mapsto(1,-1,0)$, $\lambda=-1$, image $y-z+1=0$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 165
How to find the image of a point with respect to a given plane, measured parallel to a given line?

Model Answer
Line through P parallel to the given line meets the plane at M; M is the mid-point of PP', so $P' = 2M - P$.

Ex: $(1, -2, 3)$, plane $x - y + z = 5$, parallel to $(2, 3, -6)$: $M = \left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$, $P' = \left(\frac{11}{7}, -\frac{8}{7}, \frac{9}{7}\right)$.

Grade - Low
Practice map, main concept: Sheet: J-I 50
Also used in: none

Tetrahedron and its properties

Question 166
What is a tetrahedron?

Model Answer
A pyramid with a triangular base: the solid bounded by four triangular faces, e.g. $ABCD$ with base ABC and apex D.

Grade - Medium
Practice map, main concept: none
Also used in: Question Bank: 9

Question 167
Find the number of faces, vertices and edges of a tetrahedron.

Model Answer
4 faces, 4 vertices, 6 edges, forming 3 pairs of opposite edges: AB, CD; AC, BD; AD, BC.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 168
What do you mean by lateral edges of a tetrahedron?

Model Answer
The edges joining the apex to the vertices of the base: for base ABC and apex D, the lateral edges are DA, DB, DC.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 169
What do you mean by coterminous edges of a tetrahedron?

Model Answer
Three edges meeting at a common vertex, e.g. AB, AC, AD; the volume is written with them.

Grade - Medium
Practice map, main concept: none
Also used in: Question Bank: 9

Question 170
What is the formula for the volume of a tetrahedron?

Model Answer

$$V = \frac{1}{3}(\text{area of base})(\text{height}) = \frac{1}{6}\left|\left[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}\right]\right|$$

Ex: $A(0,1,2), B(3,0,1), C(4,3,6), D(2,3,2)$: $V = 6$.

Grade - **Medium**
Practice map, main concept: DPP: 8(12); 10(10)
Also used in: Sheet: J-M 41; C-II 6 | Question Bank: 18

Question 171
How to find the height of a tetrahedron?

Model Answer
Height = projection of any lateral edge on the normal of the base: $h = |\overrightarrow{BA} \cdot \hat{n}|$, $\vec{n} = \overrightarrow{BC} \times \overrightarrow{BD}$ (or $h = \frac{3V}{\text{base area}}$).
Ex: above, base BCD: $\vec{n} \parallel (2,1,-1)$, $\overrightarrow{BA} = (-3,1,1)$, $h = \frac{6}{\sqrt{6}} = \sqrt{6}$.

Grade - **Medium**
Practice map, main concept: Sheet: C-II 6 | Question Bank: 18
Also used in: none

Question 172
What is the formula of the position vector of the centre of a tetrahedron?

Model Answer

$$\vec{g} = \frac{\vec{a} + \vec{b} + \vec{c} + \vec{d}}{4}$$

Ex: $A(0,1,2), B(3,0,1), C(4,3,6), D(2,3,2)$: centre $\left(\frac{9}{4}, \frac{7}{4}, \frac{11}{4}\right)$.

Grade - **Low**
Practice map, main concept: none
Also used in: DPP: 8(12)

Question 173
What is the point of intersection of the lines joining the mid-points of opposite edges?

Model Answer
The centre $\frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$: the mid-point of the join of $\frac{\vec{a}+\vec{b}}{2}$ and $\frac{\vec{c}+\vec{d}}{2}$ is this point, and the same for the other two pairs, so all three joins bisect each other there.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 174
What is the point of intersection of the lines joining all vertices to the centroids of the opposite faces?

Model Answer
Again the centre $\frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$; it divides each such line in the ratio 3 : 1 from the vertex, e.g. $\frac{\vec{d}+3\cdot\frac{\vec{a}+\vec{b}+\vec{c}}{3}}{4}$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 175
What do you mean by a regular tetrahedron?

Model Answer
Every face is an equilateral triangle (all 6 edges equal). Then opposite edges are perpendicular, the angle between two faces is $\cos^{-1}\frac{1}{3}$ and $V = \frac{a^3}{6\sqrt{2}}$ for edge a.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-I 18; C-II 12, 32

Parallelopiped and its properties

Question 176
What is a parallelopiped?

Model Answer
A prism with a parallelogram base: all six faces are parallelograms, opposite faces parallel and congruent. It is fixed by three coterminous edges $\vec{a}, \vec{b}, \vec{c}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 177
Find the number of faces, vertices and edges of a parallelopiped.

Model Answer
6 faces, 8 vertices, 12 edges (3 sets of 4 parallel edges).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 178
Find the number of face diagonals and body diagonals of a parallelopiped.

Model Answer
12 face diagonals (2 on each of the 6 faces) and 4 body diagonals (joining opposite vertices, e.g. O to $\vec{a}+\vec{b}+\vec{c}$).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 179
What is the formula for the volume of a parallelopiped?

Model Answer

$$V = (\text{area of base})(\text{height}) = \left|[\vec{a}\ \vec{b}\ \vec{c}]\right|$$

$\vec{a}, \vec{b}, \vec{c}$ = coterminous edges. Ex: $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$, $\hat{k}+\hat{i}$: $V = 2$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 180
How to find the height of a parallelopiped?

Model Answer

Height = projection of a lateral edge $\vec{c}$ on the normal of the base $(\vec{a}, \vec{b})$: $h = \dfrac{|\vec{c} \cdot (\vec{a} \times \vec{b})|}{|\vec{a} \times \vec{b}|} = \dfrac{|[\vec{a}\ \vec{b}\ \vec{c}]|}{|\vec{a} \times \vec{b}|}$.

Ex: above, base $\hat{i} + \hat{j}$, $\hat{j} + \hat{k}$: $\vec{a} \times \vec{b} = (1, -1, 1)$, $h = \frac{2}{\sqrt{3}}$.

Grade - Low

Practice map, main concept: DPP: 9(1)

Also used in: none

Question 181
How to find the position vector of the centre of a parallelopiped?

Model Answer

Centre = mid-point of any body diagonal (all four bisect each other there). With vertex O and coterminous edges $\vec{a}, \vec{b}, \vec{c}$: $\dfrac{\vec{a} + \vec{b} + \vec{c}}{2}$.

Ex: unit cube with edges $\hat{i}, \hat{j}, \hat{k}$ at O: centre $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$.

Grade - Low

Practice map, main concept: none

Also used in: none

STP

Question 182
Write the expression of $(\vec{a} \times \vec{b}) \cdot \vec{c}$.

Model Answer

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = [\vec{a}\ \vec{b}\ \vec{c}] = |\vec{a}||\vec{b}||\vec{c}| \sin\theta \cos\phi$$

θ = angle between $\vec{a}, \vec{b}$; ϕ = angle between $\vec{a} \times \vec{b}$ and $\vec{c}$. It is a scalar, $\pm$ the volume of the parallelopiped on $\vec{a}, \vec{b}, \vec{c}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 24, 120, 166; C-I 8 | DPP: 8(1, 10) | Question Bank: 43

Also used in: Sheet: J-II 59; J-A 14, 21 | DPP: 8(5); 9(10); 10(1, 9)

Question 183
Find the value of $(\vec{a} \times \vec{b}) \cdot \vec{c}$ if the vectors are expressed in $\hat{i}, \hat{j}, \hat{k}$ form.

Model Answer

$$[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Ex: $[(1, 2, 3)\ (0, 1, 2)\ (1, 0, 1)] = 2$.

Grade - Medium

Practice map, main concept: Sheet: J-M 161; J-A 51; C-I 1 | DPP: 8(4-5) | Question Bank: 33

Also used in: Sheet: J-I 24; J-M 5, 7, 9, 59, 64, 66, 75, 81, 179; J-A 6-7; C-II 10, 14, 26, 37 | DPP: 8(3, 7, 9); 9(8); 10(3, 11) | Question Bank: 63

Question 184
What is the effect on the value of the scalar triple product if we change the vectors in cyclic order?

Model Answer

$$[\vec{a}\ \vec{b}\ \vec{c}] = [\vec{b}\ \vec{c}\ \vec{a}] = [\vec{c}\ \vec{a}\ \vec{b}]$$

No change (cyclic property).

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 73 | DPP: 8(6)

Question 185
What is the effect on the value of the scalar triple product if we interchange two vectors?

Model Answer

$$[\vec{b}\ \vec{a}\ \vec{c}] = -[\vec{a}\ \vec{b}\ \vec{c}]$$

Only the sign changes. Hence if two vectors are equal (or parallel) the value is 0.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 23 | DPP: 8(6)

Question 186
Can dot and cross be interchanged in the scalar triple product?

Model Answer

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c})$$

Yes, keeping the cyclic order of the vectors the same.

Grade - Low
Practice map, main concept: DPP: 8(6)
Also used in: Sheet: J-II 60 | Question Bank: 43

Question 187
What is the value of the scalar triple product of $\hat{i}, \hat{j}, \hat{k}$?

Model Answer

$$[\hat{i}\ \hat{j}\ \hat{k}] = 1, \qquad [\hat{i}\ \hat{k}\ \hat{j}] = -1$$

Grade - Low
Practice map, main concept: none
Also used in: none

Question 188
What is the effect on the value of the scalar triple product if we multiply one vector by a real number k?

Model Answer

$$[k\vec{a}\ \vec{b}\ \vec{c}] = k[\vec{a}\ \vec{b}\ \vec{c}]$$

The value is multiplied by k; e.g. $[2\vec{a}\ 3\vec{b}\ \vec{c}] = 6[\vec{a}\ \vec{b}\ \vec{c}]$.

Grade - Low
Practice map, main concept: Sheet: J-I 23
Also used in: DPP: 8(3)

Question 189
How to find the scalar triple product of three vectors if they are expressed as linear combinations of three other vectors?

Model Answer
If $\vec{u} = x_1\vec{a} + y_1\vec{b} + z_1\vec{c}$, $\vec{v} = x_2\vec{a} + \ldots$, $\vec{w} = x_3\vec{a} + \ldots$, then

$$[\vec{u}\ \vec{v}\ \vec{w}] = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} [\vec{a}\ \vec{b}\ \vec{c}]$$

Ex: $[\vec{a}+\vec{b}\ \ \vec{a}-\vec{b}\ \ \vec{c}] = -2[\vec{a}\ \vec{b}\ \vec{c}]$.

Grade - Medium
Practice map, main concept: Sheet: J-I 18; J-M 63; C-II 10 | DPP: 10(6)
Also used in: Sheet: J-I 20; J-M 21; J-A 12, 56; C-II 16

Question 190
How to find the scalar triple product of three vectors in terms of the dot products between them?

Model Answer

$$[\vec{a}\ \vec{b}\ \vec{c}]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$$

Ex: unit vectors with mutual angles 60°: $[\vec{a}\ \vec{b}\ \vec{c}]^2 = \frac{1}{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 19, 21; J-A 40; C-I 6
Also used in: none

Question 191
How to find the volume of a parallelopiped using the scalar triple product?

Model Answer

$$V = \left|[\vec{a}\ \vec{b}\ \vec{c}]\right|$$

$\vec{a}, \vec{b}, \vec{c}$ = coterminous edges (from vertices: $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}$ from one vertex). Ex: $2\hat{i}$, $3\hat{j}$, $\hat{i}+\hat{j}+4\hat{k}$: $V = 24$.

Grade - Medium
Practice map, main concept: Sheet: J-A 6; C-I 17; C-II 26, 43 | DPP: 8(3); 10(3)
Also used in: Sheet: J-I 21; J-II 36; J-M 63; C-I 3, 20 | DPP: 9(1)

Question 192
How to find the volume of a tetrahedron using the scalar triple product?

Model Answer

$$V = \frac{1}{6}\left|[\vec{b}-\vec{a}\ \ \vec{c}-\vec{a}\ \ \vec{d}-\vec{a}]\right|$$

Ex: vertices $O, (1,0,0), (0,2,0), (0,0,3)$: $V = 1$.

Grade - Low
Practice map, main concept: Sheet: J-II 20-21; C-II 12, 31, 39
Also used in: Sheet: J-II 41; C-II 9 | DPP: 9(5)

Question 193
What is the value of $[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}]$?

Model Answer

$$[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}] = 2[\vec{a}\ \vec{b}\ \vec{c}]$$

(Coefficient determinant $= 2$.) Similarly $[\vec{a}-\vec{b}\ \ \vec{b}-\vec{c}\ \ \vec{c}-\vec{a}] = 0$.

Grade - Medium
Practice map, main concept: Sheet: J-A 12; C-I 3; C-II 27 | DPP: 9(5)
Also used in: none

Coplanarity of three vectors

Question 194
What is the condition for three vectors to be coplanar?

Model Answer

$$[\vec{a}\ \vec{b}\ \vec{c}] = 0$$

Equivalently one of them is a linear combination of the other two (they are linearly dependent). Ex: $\hat{i}+\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+3\hat{k}$, $2\hat{i}+3\hat{j}+4\hat{k}$: determinant 0, coplanar.

Grade - Medium
Practice map, main concept: Sheet: J-I 20, 24; J-II 66; J-M 7, 21, 66, 75, 81; J-A 7, 56; C-I 18 | DPP: 7(4); 8(7)
Also used in: Sheet: J-I 54; J-II 42, 70; C-II 14 | DPP: 8(2); 10(6)

Question 195
How to check whether four points are coplanar?

Model Answer

$$A, B, C, D \text{ coplanar} \iff \left[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}\right] = 0$$

Ex: $(1,0,0), (0,1,0), (0,0,1), (k,k,k)$ are coplanar iff $k = \frac{1}{3}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 5, 59, 64, 73; J-A 48
Also used in: Sheet: C-II 16

Question 196
How to check whether two lines are coplanar?

Model Answer

$$\vec{r} = \vec{a}_1 + \lambda\vec{b}_1,\ \vec{r} = \vec{a}_2 + \mu\vec{b}_2 \text{ coplanar} \iff [\vec{a}_2 - \vec{a}_1 \quad \vec{b}_1 \quad \vec{b}_2] = 0$$

(Parallel lines are always coplanar; non-parallel coplanar lines intersect.) Ex: $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$, $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{1}$: determinant 0, they meet at $(-1, -1, -1)$.

Grade - **Medium**
Practice map, main concept: Sheet: J-I 39; J-II 30; J-M 107, 111; J-A 5, 10, 53 | DPP: 1(8); 7(1, 6)
Also used in: Sheet: J-II 43, 64, 67; J-M 102; C-I 22

VTP

Question 197
What is the direction of $(\vec{a} \times \vec{b}) \times \vec{c}$?

Model Answer
It is perpendicular to $\vec{a} \times \vec{b}$, so it lies in the plane of $\vec{a}$ and $\vec{b}$, and it is perpendicular to $\vec{c}$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 198
What is the formula of $(\vec{a} \times \vec{b}) \times \vec{c}$?

Model Answer

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}, \qquad \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

In general $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$. Ex: $(\hat{i} \times \hat{j}) \times \hat{i} = \hat{j}$.

Grade - **Medium**
Practice map, main concept: Sheet: J-I 30-33; J-II 22, 60; J-M 10-11, 84, 135, 167; C-I 23; C-II 13, 46 | DPP: 9(7); 10(8, 11, 13) | Question Bank: 2, 59, 63
Also used in: Sheet: J-I 11, 26, 52-53; J-M 3-4, 6, 179; J-A 13, 57; C-I 4, 9; C-II 19, 37-38 | DPP: 5(2); 9(6)

Question 199
How to assume a vector coplanar with $\vec{p}$ and $\vec{q}$ and perpendicular to $\vec{r}$?

Model Answer

$$\vec{v} = \lambda\,\vec{r} \times (\vec{p} \times \vec{q}) = \lambda\,[(\vec{r} \cdot \vec{q})\vec{p} - (\vec{r} \cdot \vec{p})\vec{q}]$$

Ex: $\vec{p} = \hat{i} + \hat{j}$, $\vec{q} = \hat{j} + \hat{k}$, $\vec{r} = \hat{i} + \hat{j} + \hat{k}$: $\vec{v} = \lambda(\hat{i} - \hat{k})$.

Grade - **Medium**
Practice map, main concept: Sheet: J-I 11, 26; J-M 4; J-A 13; C-I 24 | DPP: 9(9)
Also used in: none

Question 200
What is the value of $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}]$?

Model Answer

$$[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}] = [\vec{a}\ \vec{b}\ \vec{c}]^2$$

Ex: for $\hat{i},\hat{j},\hat{k}$: $[\hat{k}\ \hat{i}\ \hat{j}] = 1 = 1^2$.

Grade - Low
Practice map, main concept: Sheet: J-I 27 | DPP: 9(10)
Also used in: none

Question 201
What is the result of $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d})$?

Model Answer

$$(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = \begin{vmatrix}\vec{a}\cdot\vec{c} & \vec{a}\cdot\vec{d}\\ \vec{b}\cdot\vec{c} & \vec{b}\cdot\vec{d}\end{vmatrix} = (\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d}) - (\vec{a}\cdot\vec{d})(\vec{b}\cdot\vec{c})$$

Ex: $\vec{c}=\vec{a}$, $\vec{d}=\vec{b}$: $|\vec{a}\times\vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a}\cdot\vec{b})^2$ (Lagrange identity).

Grade - Medium
Practice map, main concept: Sheet: J-I 25; J-M 3; C-I 12
Also used in: Sheet: J-II 61, 69

Question 202
How to apply vector triple product results if more than three vectors are given?

Model Answer
Treat a cross product such as $\vec{a}\times\vec{b}$ as a single vector and apply the three-vector formula; for a dot of two cross products use the item above.

$$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = [\vec{a}\ \vec{b}\ \vec{d}]\vec{c} - [\vec{a}\ \vec{b}\ \vec{c}]\vec{d} = [\vec{a}\ \vec{c}\ \vec{d}]\vec{b} - [\vec{b}\ \vec{c}\ \vec{d}]\vec{a}$$

Equating the two forms: $[\vec{b}\,\vec{c}\,\vec{d}]\vec{a} - [\vec{a}\,\vec{c}\,\vec{d}]\vec{b} + [\vec{a}\,\vec{b}\,\vec{d}]\vec{c} - [\vec{a}\,\vec{b}\,\vec{c}]\vec{d} = \vec{0}$ (any four vectors in space).

Grade - Low
Practice map, main concept: Sheet: J-II 61; C-II 14
Also used in: none

Linear Combination theorem in space

Question 203
Explain the linear combination theorem in space.

Model Answer
If $\vec{a},\vec{b},\vec{c}$ are non-coplanar, every vector $\vec{r}$ in space can be written uniquely as $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$, with

$$x = \frac{[\vec{r}\ \vec{b}\ \vec{c}]}{[\vec{a}\ \vec{b}\ \vec{c}]},\quad y = \frac{[\vec{a}\ \vec{r}\ \vec{c}]}{[\vec{a}\ \vec{b}\ \vec{c}]},\quad z = \frac{[\vec{a}\ \vec{b}\ \vec{r}]}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

Ex: $2\hat{i}+3\hat{j}+4\hat{k}$ with $\hat{i}$, $\hat{i}+\hat{j}$, $\hat{i}+\hat{j}+\hat{k}$: $x=-1,\ y=-1,\ z=4$.

Grade - Medium
Practice map, main concept: Sheet: J-A 14, 17, 52; C-II 16-17 | DPP: 10(9)
Also used in: Sheet: C-II 44

Question 204
What is the condition to express any vector as a linear combination of three vectors?

Model Answer

$$[\vec{a}\ \vec{b}\ \vec{c}] \neq 0$$

The three vectors must be non-coplanar (linearly independent). Ex: $\hat{i}+\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+3\hat{k}$, $2\hat{i}+3\hat{j}+4\hat{k}$ are coplanar, so $\hat{k}$ cannot be written with them.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 205
How to express $\vec{r}$ in terms of $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$ and $\vec{c}\times\vec{a}$?

Model Answer

$$\vec{r} = \frac{(\vec{r}\cdot\vec{a})(\vec{b}\times\vec{c}) + (\vec{r}\cdot\vec{b})(\vec{c}\times\vec{a}) + (\vec{r}\cdot\vec{c})(\vec{a}\times\vec{b})}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

(Take $\vec{r} = x(\vec{b}\times\vec{c}) + y(\vec{c}\times\vec{a}) + z(\vec{a}\times\vec{b})$ and dot with $\vec{a},\vec{b},\vec{c}$; needs $[\vec{a}\ \vec{b}\ \vec{c}] \neq 0$.)

Grade - Low
Practice map, main concept: DPP: 10(1)
Also used in: none

Vector Equation

Question 206
What to keep in mind while solving a vector equation?

Model Answer
Do not introduce new vectors; isolate the unknown vector; take dot or cross product of both sides with a known vector, then use the vector triple product formula.
Ex: $\vec{r}\times\vec{b} = \vec{a}\times\vec{b}$, $\vec{r}\cdot\vec{c} = 0 \Rightarrow (\vec{r}-\vec{a})\times\vec{b} = \vec{0}$, $\vec{r} = \vec{a} + t\vec{b}$, $t = -\dfrac{\vec{a}\cdot\vec{c}}{\vec{b}\cdot\vec{c}}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 52-53; J-II 59; J-M 1, 6, 18, 20, 74, 139, 164, 171, 178-180; J-A 44, 57; C-I 4, 9; C-II 19, 37-38 | DPP: 5(1); 9(6) | Question Bank: 66
Also used in: Sheet: J-I 28; J-M 134; C-II 39 | Question Bank: 2, 63-64

Line of Greatest slope

Question 207
How to find the line of greatest slope?

Model Answer

$$\vec{d} = (\vec{n}_H \times \vec{n}_G) \times \vec{n}_G$$

It is the line in the given plane G perpendicular to the line of intersection of G with the horizontal plane H; a given point fixes it.
Ex: $H: 4x-3y+7z=0$, $G: 2x+y-5z=0$, through $(2,1,1)$: DRs $3,-1,1$, line $\frac{x-2}{3} = \frac{y-1}{-1} = \frac{z-1}{1}$.

Grade - Low
Practice map, main concept: Sheet: J-II 47

Also used in: none

Miscellaneous

Question 208

How to find a line through a given point (or with a given direction) that intersects two given lines, and the distance of a point from a line measured parallel to another line?

Model Answer

Take general points $P(t)$ on L_1 and $Q(s)$ on L_2. Through a point A: A, P, Q collinear ($\overrightarrow{AP} \parallel \overrightarrow{AQ}$). Given direction $\vec{d}$: $\overrightarrow{PQ} \parallel \vec{d}$. Solve for t, s.

Same idea: distance of A from a line measured parallel to another line $\vec{d}$: take $P(t)$ on the line with $\overrightarrow{AP} \parallel \vec{d}$; distance $= AP$.

Ex: through O meeting $\vec{r} = (1,0,1) + t\hat{j}$ and $\vec{r} = (0,1,2) + s\hat{i}$: $t = \frac{1}{2}$, $s = 2$, line $\frac{x}{2} = \frac{y}{1} = \frac{z}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 35; J-M 30, 87, 142, 153, 156, 189, 191; J-A 33; C-II 45 | Question Bank: 11, 42, 67

Also used in: none

Question 209

How to use vectors for geometric proofs and maximum / minimum problems?

Model Answer

Expand squares of magnitudes with dot products and use $|\vec{v}|^2 \geq 0$, $|\cos\theta| \leq 1$, $|\sin\theta| \leq 1$ or AM-GM.

$$|\vec{a} \cdot \vec{b}| \leq |\vec{a}||\vec{b}|, \qquad |\vec{a} \times \vec{b}| \leq |\vec{a}||\vec{b}|$$

Equality: $\vec{a} \parallel \vec{b}$ in the first, $\vec{a} \perp \vec{b}$ in the second. Ex: max of $x + 2y + 2z$ on $x^2 + y^2 + z^2 = 1$ is $|(1,2,2)| = 3$.

Ex: unit $\vec{a}, \vec{b}, \vec{c}$: $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9 - |\vec{a} + \vec{b} + \vec{c}|^2 \leq 9$.

Grade - Medium

Practice map, main concept: Sheet: J-I 8; J-II 8; J-A 1; C-I 7; C-II 42 | DPP: 5(11)

Also used in: Sheet: J-II 7, 12; J-M 24, 71, 175; C-I 8; C-II 29, 33 | DPP: 8(16); 9(10) | Question Bank: 39

Question 210

If $p\overrightarrow{OA} + q\overrightarrow{OB} + r\overrightarrow{OC} = \vec{0}$ for a point O inside $\triangle ABC$, how are the areas of the sub-triangles related?

Model Answer

$$\Delta BOC : \Delta COA : \Delta AOB = p : q : r, \qquad \frac{\Delta BOC}{\Delta ABC} = \frac{p}{p+q+r}$$

Proof: take the cross product of the relation with $\overrightarrow{OA}$, $\overrightarrow{OB}$ ($p, q, r > 0$).

Ex: $2\overrightarrow{OA} + 3\overrightarrow{OB} + 5\overrightarrow{OC} = \vec{0} \Rightarrow \Delta BOC = \frac{1}{5}\Delta ABC$.

Grade - Low

Practice map, main concept: Sheet: J-II 14 | DPP: 5(9)

Also used in: Sheet: J-II 69